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# A digital solution for CPS-based machining path optimization for CNC systems

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**Abstract**

In recent years, the deep integration of advanced information technology and advanced manufacturing technology has gradually become one of the main ways to achieve smart manufacturing. The computer numerical control (CNC) system is the basic equipment for machining and manufacturing, and the quality and efficiency of the system's machining are the basis for supporting and ensuring smart manufacturing. However, the G-code used in CNC machining is usually generated with computer-aided manufacturing (CAM) according to a static model, and its tool path is relatively rough, with uneven adjacent paths and bad points in the path causing machining defects. To solve these problems, a modeling approach combining the basic elements of the intelligent CNC system with the human-cyber-physical system (HCPS) model is proposed, and a digital solution for tool path optimization is further proposed, integrating the redesign process of CAM tool path into cyber application. In addition, the process of tool path optimization is processed in steps, and a pipelined processing flow is established to accelerate the optimization process. Finally, the effectiveness of the proposed method is demonstrated using an example of process file optimization for a pentagram convex rib model.

**Keywords** CNC system · Intelligent elements · Cyber-physical systems · Process analysis · Path optimization

## Presentation of the problem

The new generation of artificial intelligence technologies and advances in information technology are driving the development of CNC systems in the direction of digitization and intelligence. In particular, the integration of technologies such as the digital twin, the Internet of Things, big data, and edge computing into a CNC system can make such a system more functional and extend its structure (Martinov et al., 2020; Wang et al., 2023). At the same time, technological advances have led to changes in industrial production methods, with competition in the economic and business environment leading more workshops and factories to adopt more intelligent machining and production methods (González Rodríguez et al., 2020; Imad et al., 2022). As the brain of machining and manufacturing equipment, the accuracy of CNC system instructions directly affects the machining quality. Therefore, it is necessary to study intelligent optimization methods and approaches for machining instructions in CNC systems.

Currently, automated manufacturing relies on data transmission between CNC machine tools and CAM in industrial practice. The CAM system plans the machining process of the CNC and fundamentally determines the quality and efficiency of machining (Lynn et al., 2020). However, the machining program directly generated by the CAM system only pays attention to the continuity of the position and does not consider the situation of the movement, so the tool path is relatively rough and includes some negative feature points and feature edges. As shown in Fig. 1, the sharp corners of the tool path, uneven adjacent paths, and uneven distribution of points will all have a negative impact on machining.

In previous work, the data input from the CAM system was only in the program design stage and was not allowed

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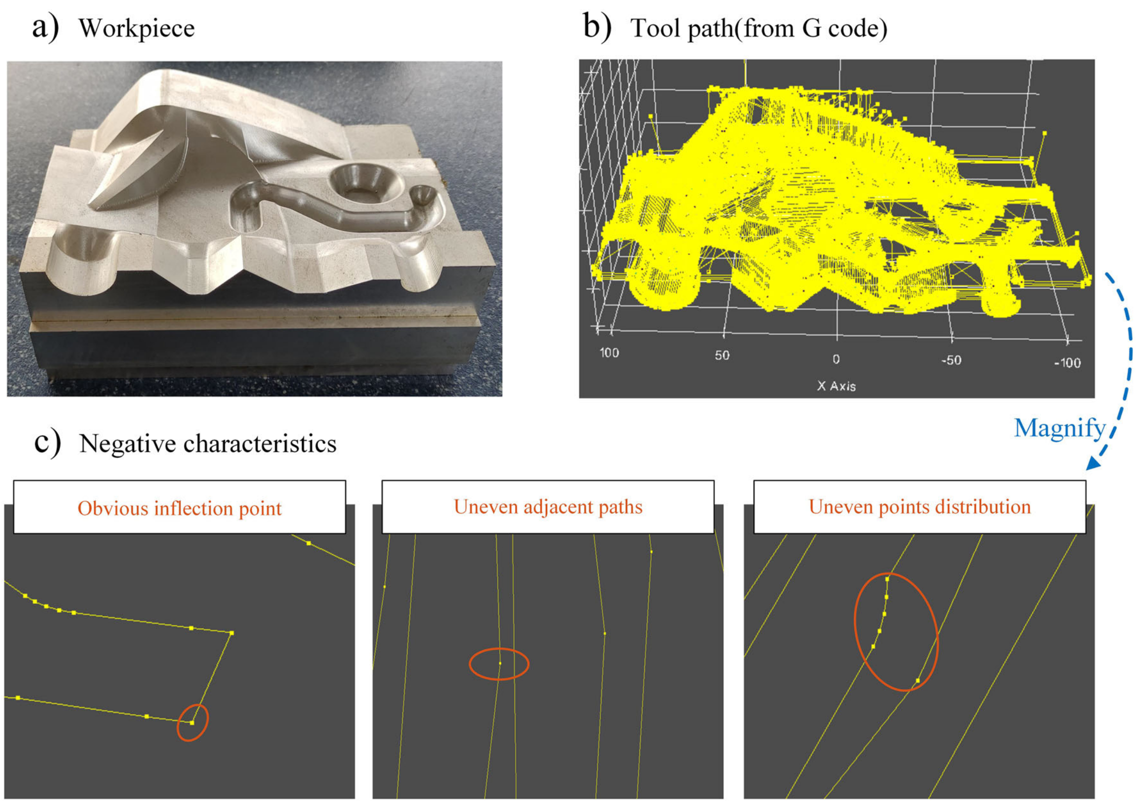



**Fig. 1** Visualization examples of point defects in different situations in the original machining program generated by the CAM system

to make process decisions based on the data collected during the machining process. The data transmission between the CAM and the machine tool controller was unidirectional, as shown in Fig. 2a. The optimization of negative feature points usually relies on the participation of CNC programmers. This processing method requires considerable manual effort and is restricted by objective factors such as the operator's skill level and workpiece shape.

Therefore, it is necessary to consider adopting more efficient and intelligent methods to improve production efficiency and production quality. To solve this problem, we established a new working method, as shown in Fig. 2b. A closed loop of data transmission is established between the CAM system and the machine tool controller, and the optimization process is encapsulated in the cyber-physical-system (CPS) unit in the form of a cyber application integrated with the machine tool controller.

To summarize, considering the integration of a CAM system and machine tool controller, this paper studies an intelligent optimization method for the machining path of a CNC system based on the HCPS concept. The contributions of this paper can be summarized as follows.

(1) A scheme for the intelligent extension of CNC systems is proposed, establishing a model that combines the basic elements of intelligent CNC systems with HCPS. The concept and implementation of abstracting the intelligent elements in the cyber system into a cyber agent are proposed, providing intellectual support for the CNC system.
(2) A digital solution for CNC system tool path optimization based on HCPS is proposed, and a digital twin model of the CNC system is established for machining path simulation.
(3) The path optimization process based on HCPS has been divided into stages, establishing a pipelined processing flow to improve the efficiency of path optimization.
(4) In the path optimization stage, a process analysis application is established. A path hierarchical decomposition algorithm is proposed to help locate defect paths. With the assistance of deep learning methods for identification and analysis, an adaptive defect path optimization method based on clustering segmentation is proposed, and in combination with the process above, a visualization application is established.

The overall structure of this paper consists of six sections. The problem is described in "Presentation of the problem" section, and the current situation of the problem and the proposed new solutions are briefly described. In "Related works" section, an overview of work related to HCPS, digital twins, and CNC machining path optimization for CNC systems is

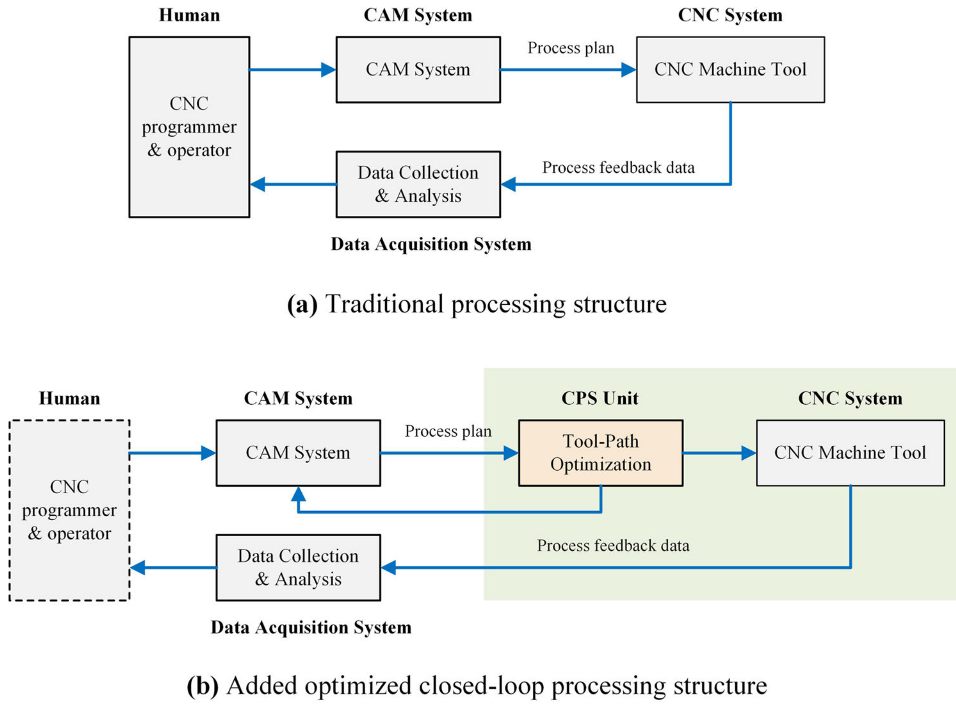



**Fig. 2** Comparison between the structure of traditional processing method (**a**) and the processing method proposed in this article (**b**)

given, and research gaps are analyzed. In "Intelligent scheme of the CNC system" section, a proposed HCPS model of the intelligent CNC system and the digital scheme of path optimization are introduced. "Pipeline working mode of path optimization" section presents the pipelined path optimization process. "Experimental study" section presents a case study of path optimization of the intelligent CNC system based on HCPS. "Conclusion" section summarizes the conclusions and future research directions.

## Related works

This section reviews the relevant research on intelligent CNC systems and path optimization of CNC systems, summarizes their development status, analyzes deficiencies, and explains the theoretical and practical value of the research content.

### Research on the intelligentization of CNC system

CNC systems play an important role in the manufacturing industry, and their development has received widespread attention. Since the concept of Industry 4.0 was proposed, the intellectualization and digitalization of the CNC system have been constantly mentioned and studied (Tong et al., 2020). The concept of intelligent expansion of the CNC system focuses on the CNC system based on the HCPS and the CNC system based on digital twins. In essence, they all realize the deep integration of the information world and the physical world.

(1) *Intelligent CNC system based on HCPS* HCPS is a technical concept in the new generation of intelligent manufacturing. It assists human beings in perception, analysis, and decision-making functions by adding a cyber system between human and physical systems (Zhou et al., 2018). Zhou et al. (2019b) elaborated on the evolution of HCPS for intelligent manufacturing and analyzed the characteristics of HCPS from the perspective of systems and technology. Chen et al. (2019) discussed the three stages of machine tool development: manually operated machine tools and numerical control machine tools, to intelligent machine tools. They introduced the intelligent practice of intelligent machine tools. Liu et al. (2018) proposed a systematic development method of information physics machine tools, which allows machine tools, processing processes, and intelligent algorithms to be integrated through the network. Liu et al. (2019b) proposed a cyber-physical machine tools (CPMT) platform based on OPC UA and MTConnect to realize the communication between application programs and machine tools.

(2) *Intelligent CNC system based on digital twin* Digital twinning technology focuses on mapping information to the physical object model and controlling and manipulating the physical object model by studying and operating the information model. Its use helps to solve the problems of complex and variable faults in manufacturing (Deebak & Al-Turjman, 2022; Wang et al., 2022), human–machine interaction and visualization in cyber-physical systems (Fan et al., 2021; Leng et al.,

2021; Niermann et al., 2023), fault avoidance in irreversible design methods (Cao et al., 2022; Gopal et al., 2023; Liu et al., 2022a), and predictive maintenance of manufacturing systems and tools (Luo et al., 2020; Van Dinter et al., 2022). Wei et al. (2021) studied a method for maintaining the consistency of digital twins of CNC machine tools and demonstrated the feasibility of the method by analyzing the wear performance degradation of rolling guide rails. He et al. (2019) integrated virtual modeling, process monitoring, diagnosis, and optimal control in automation systems and established a data-driven digital twin system based on this analysis. Zhou et al. (2021) established a five-dimensional digital twin model and constructed a digital-twin-based optimization strategy and experimental verification method for the processing efficiency and aerodynamic performance of the centrifugal impeller. Yu et al. (2023) established a digital twin model of a CNC system driven by edge intelligence technology and discussed the deployment method of the model.

In industry, many leading companies have also developed digital products or related projects, such as the MindSphere digital twin application of Siemens (Pauli et al., 2021), DMG MORI's integrated digital manufacturing solution based on their operating system platform (CELOS), and FANUC's automation, digitization, and intelligent solutions.

## CNC machining path optimization

CNC machining is a processing method widely used in mechanical manufacturing systems (Zhou et al., 2019a). The machine program of the CNC system is usually generated by a CAM system using computer-aided design (CAD) information. When inputting information into a CNC system, those data are converted into a G-code defined by digital codes such as G, T, M, S, and F (ISO 6983) (Latif et al., 2021). The CNC controller reads the G-code in the command, interprets it, and finally uses the cutting tool to interpolate the code into the workpiece.

In the above steps, it is necessary to formulate corresponding algorithm rules to constrain physical elements such as the cutting direction, force, and length of the tool. These elements are the core of the CNC machining process. However, the G-code generated directly from the initial model usually features machining path defects, and the root cause lies in the unreasonable trajectory of the original G-code. For example, unevenness of the trajectory in the transverse direction of the tool path will cause uneven tool marks on the surface, and unevenness of the path will cause vibration of the feed axis, which will react to the surface of the workpiece and also form surface processing defects (Guo et al., 2022). The CNC system machining path optimization technology is a comprehensive application technology to improve the machining accuracy, machining efficiency, and surface quality of parts. Path smoothing is an important technical avenue for surface machining optimization.

Hu et al. (2022) proposed a method for identifying tool path feature points using deep learning technology. By converting tool path point features into images for visual processing, this method avoids relying on manual processing and improves automation efficiency. Dittrich and Uhlich (2020) investigated a method for automatic process optimization for 5-axis milling using machine learning methods to predict resulting shape errors and adjust toolpaths. Xiao et al. (2021) optimized the cutting parameters of the machining center with the particle swarm optimization algorithm and improved genetic algorithm to realize energy savings and low-cost CNC machining. Hatem et al. (2021) applied the high-performance ant colony algorithm and open system architecture control to CAD/CAM to realize the optimization of tool paths and shorten the manufacturing time of instruction sequences. Chu et al. (2020) used cubic curves to generate toolpaths and searched for optimal curve control points and the resulting toolpaths through an electromagnetic-like mechanism optimization algorithm, thereby improving the computational efficiency of toolpath generation and the machined surface finish. Herraz et al. (2021) first divided the surface into regions with similar local geometric properties; based on this division, an unsupervised clustering algorithm was used, and a tool path planning algorithm was applied to each region to optimize the path and define the optimal processing direction. Zhao et al. (2021) proposed a vibration-error-based trajectory planning method for machine tools, aiming at given the optimal location and processing time of the machining path and evaluating typical machining using the average value and fluctuation of the vibration error in the entire trajectory as an index for vibration errors in the path. In addition, path planning not only plays a crucial role in CNC machining but also plays a crucial role in other fields. Liu et al. (2020) proposed an optimal path planning system for size detection of automotive bodies, which combines probe rotation and effective collision detection to generate collision-free paths automatically. Liu et al. (2022b) proposed a novel path planning method that combines measurement uncertainty to optimize the observation points for robot-free surface detection, significantly improving the scanning accuracy of crucial measurement points.

## Research gaps

Research on related work has shown that digital and intelligent manufacturing technologies can implement intelligent control strategies promptly in complex environments, catering to today's complex and demanding manufacturing systems. In this context, both academia and the industry are highly concerned about the intelligent process of CNC systems. However, most research work is only theoretical or experimental (Liu et al., 2021), and there are still gaps and limitations in current research and practice.

(1) There is a need for more integrated intelligent CNC system models that consider the execution process, and the problem of implementing tool path optimization of CNC systems through digital technology needs to be improved.
(2) In the current manufacturing mode, the design of workpieces is separated from the actual machining process, and there needs to be more integration between CAM systems and CNC process chains.
(3) Previous research focused on tool path planning methods used after G-code is generated, ignoring the hidden value of the G-code itself.

In response to the research gaps and problems mentioned above, this paper proposes an intelligent CNC system framework based on HCPS and a digital solution for path optimization. In the proposed intellectual CNC system framework, the functional components of the CNC system are modelled as intelligent elements to reveal the execution process of the intelligent CNC system, and the intelligent elements in the Cyber system are abstracted as cyber agents to provide intelligent support for the CNC system. The redesign function of the CAM system was integrated through Cyber application, thus establishing a data transmission path between CAM and the machine tool controller. On this basis, the tool path defects were located by analyzing the tool path information in the original G-code, and the optimization of the tool path was achieved in the form of layer-by-layer decomposition.

## Intelligent scheme of the CNC system

This section analyzes the motivation for intelligent CNC systems. It proposes a solution for building an intelligent CNC system, including the HCPS abstract model and architecture of the intelligent CNC system. It further introduces the modeling process of intelligent CNC systems and the digital solution of path optimization, providing a practical reference for implementing intelligent CNC systems.

### Motivation of CNC system intelligentization

The intelligence of CNC systems is a response to the increasing complexity and efficiency demands of the modern manufacturing environment. As shown in Fig. 3, traditional CNC systems mainly consist of CNC controllers, actuators and machine tools that perform closed-loop interpolation and servo control processes, with problems and faults relying mainly on manual detection and handling. It is worth noting that the new generation of intelligent CNC systems added a computer, digital twin, closed loop of cutting process feedback and optimization for real-time simulation and prediction of the machine tool's operating status and data to optimize the production process, early warning or troubleshooting.

Intelligent CNC systems achieve automation and intelligence of manufacturing processes by introducing digital twins and artificial intelligence technology. As a virtual copy of solid machine tools, the digital twin realizes real-time simulation and machine tool operation status prediction. Introducing deep learning, machine learning, reinforcement learning and other technologies can help CNC systems better understand and control the processing process, improve processing efficiency and quality, reduce manual intervention, and achieve higher intelligence and automation. In the experimental study later in this paper, we show the application of digital twin technology, the DBscan clustering method in machine learning, and the convolutional neural network (CNN) method in deep learning in the case of tool path optimization.

Although digital technology has been dramatically developed, design and manufacturing are usually at different stages in the current manufacturing mode. The design of the workpiece is separated from the actual processing process, and the tool path defects of the workpiece can only be fed back to the design process after the negative results are observed in the manufacturing process. The redesign of the workpiece model, the workload of adjustment and improvement, is heavy and requires rich experience and a high level of domain knowledge.

The new workflow proposed in this paper is shown in Fig. 4. The previous machining process is directly used for machining and manufacturing after the CAM system generates the original tool path. The new workflow added in the blue dashed frame is the leading work content of this paper. The redesign function of the CAM system is integrated, and the data communication bridge between the CAM system and CNC is established.

### Abstract model and architecture

According to the requirement analysis of intelligent CNC systems provided in "Motivation of CNC system intelligentization" section, we propose an intelligent expansion scheme for CNC systems, as shown in Fig. 5. We maintain that the intelligent expansion of CNC systems needs to create reliable interconnections and interoperability between CNC systems and field equipment. Under this background, the proposed scheme has the characteristics of virtual realization of the

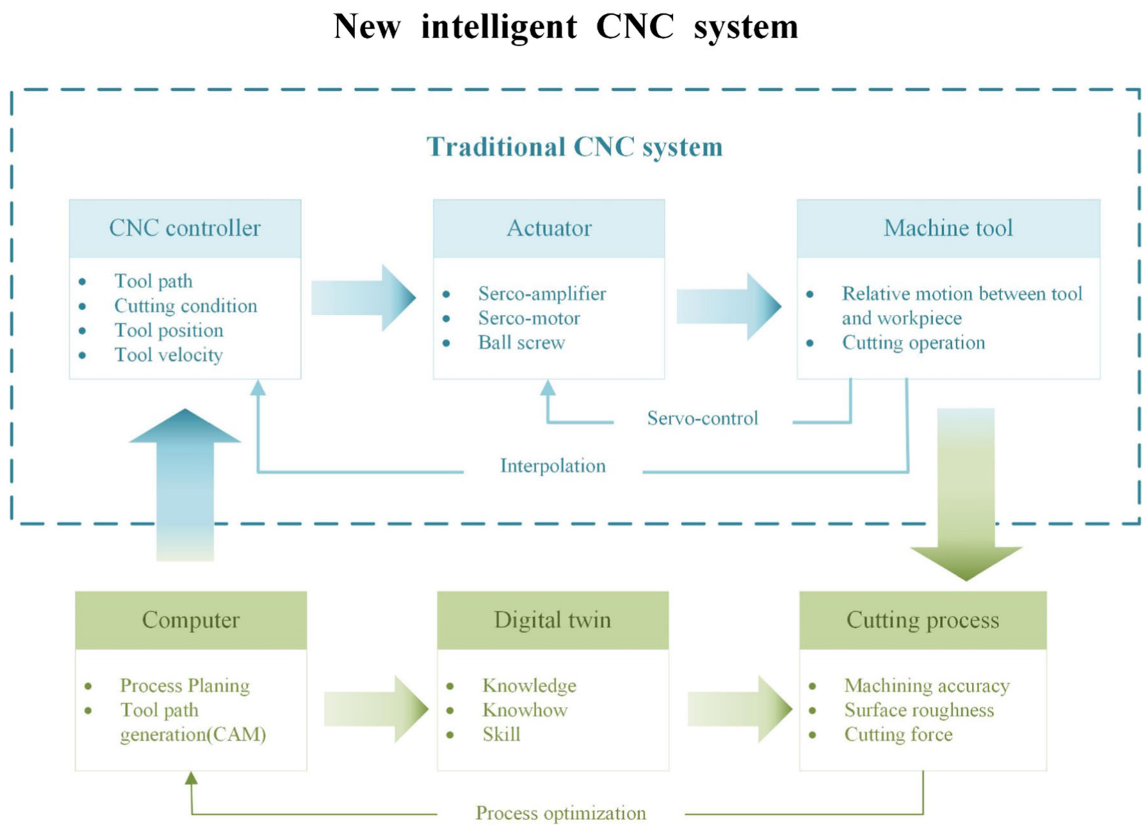



**Fig. 3** Comparative of traditional CNC system and new intelligent CNC system

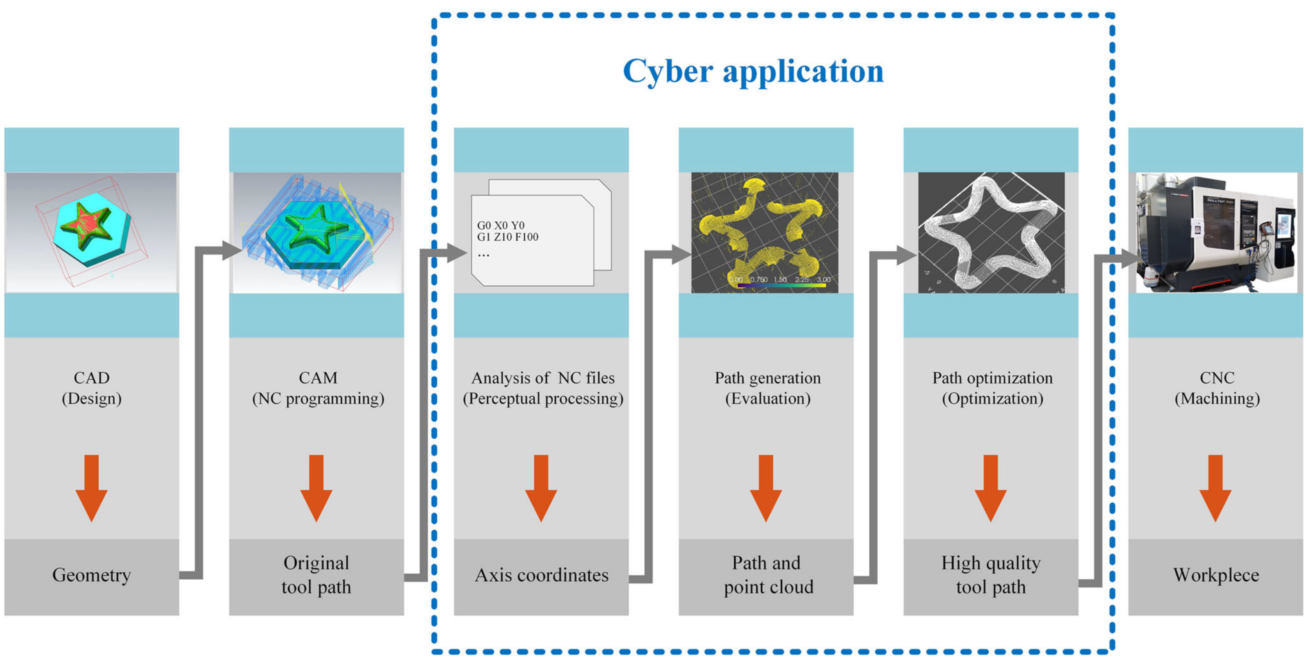



**Fig. 4** Established a bridge for information transfer between CAD/CAM and machine controller through cyber application

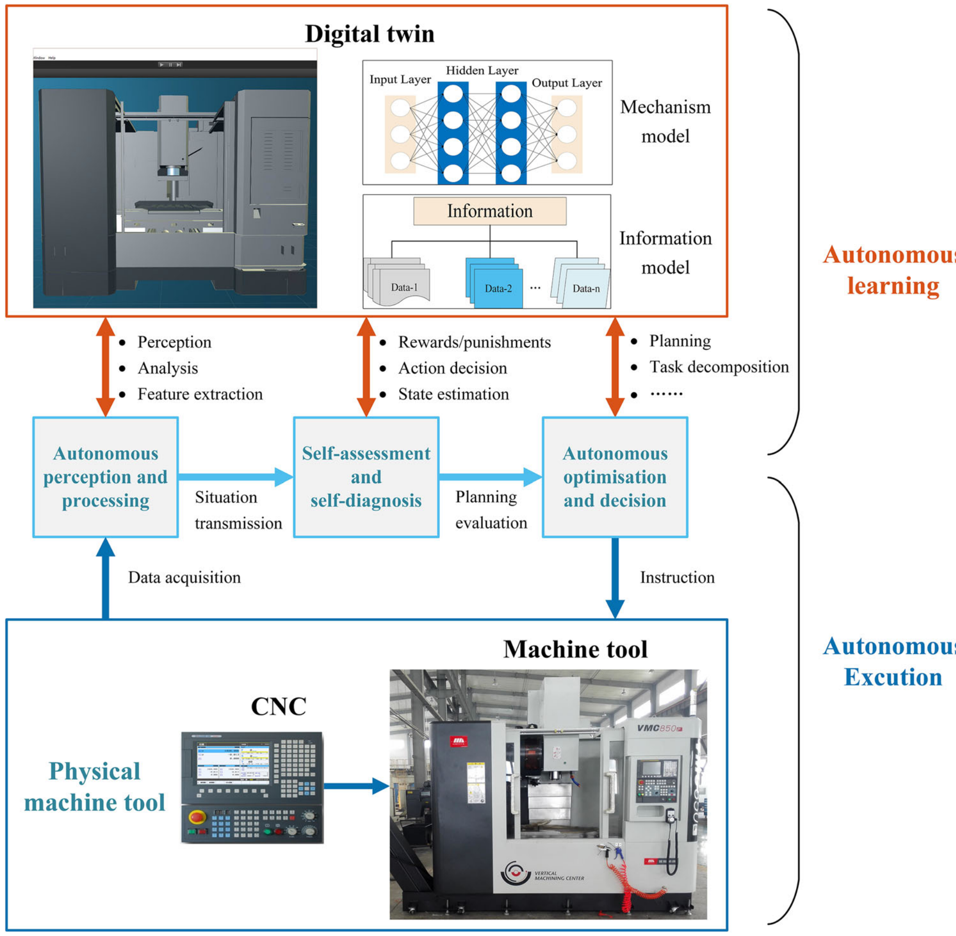



**Fig. 5** An intelligent expansion scheme for traditional CNC systems

physical world, autonomous perception and processing of input, autonomous evaluation of processing and autonomous diagnosis of faults, and adaptive decision-making and optimization.

Based on the above scheme in Fig. 5, an abstract model of intelligent CNC systems is further proposed as follows. This model provides a necessary means to promote the development of numerical control technology theory by establishing a model to describe and abstract the working process of the numerical control system. As a typical HCPS, the CNC system contains functional components with their functions and characteristics. Therefore, the HCPS abstract model of the CNC system can be obtained by summarizing the abstract model based on its functional components. As shown in Fig. 6, we summarized the whole intelligent numerical control system into human-cyber-physical areas and established an abstract functional structure model for each area. The connection and direction of the arrows between the models represent the interaction between the modules.

- *Perceptual processing* The perceptual processing module contains the algorithm processing of the CAM system's input and the sensor module's input data. The main function of this module is the perceptual calculation of the characteristics of the system input. Including distance, shape, direction, surface features, and dynamic properties.
- *Assessment and diagnosis* The assessment and diagnosis module evaluates the system elements and judges the likes and dislikes of the system, rewards and punishments of the behavior, the importance of features, and the possibility of prediction. It reminds or diagnoses the faults that will or have occurred.
- *Optimization and decision* The optimization and decision module optimizes the results that are evaluated as poor and select the best decision from many actions to generate the following work plan, such as the decision of interpolation, trajectory planning, prediction, and maintenance.
- *World model* Responsible for real-time interaction with the above modules. The world model contains the persistence of the CNC system's data and the understanding of the ontology state. The data is further extracted into knowledge in the world model module to provide the CNC system's cognition, evaluation, and prediction of the current state.
- *Actuator* The actuator module is the execution and output of the system. This module serves as the executor of the

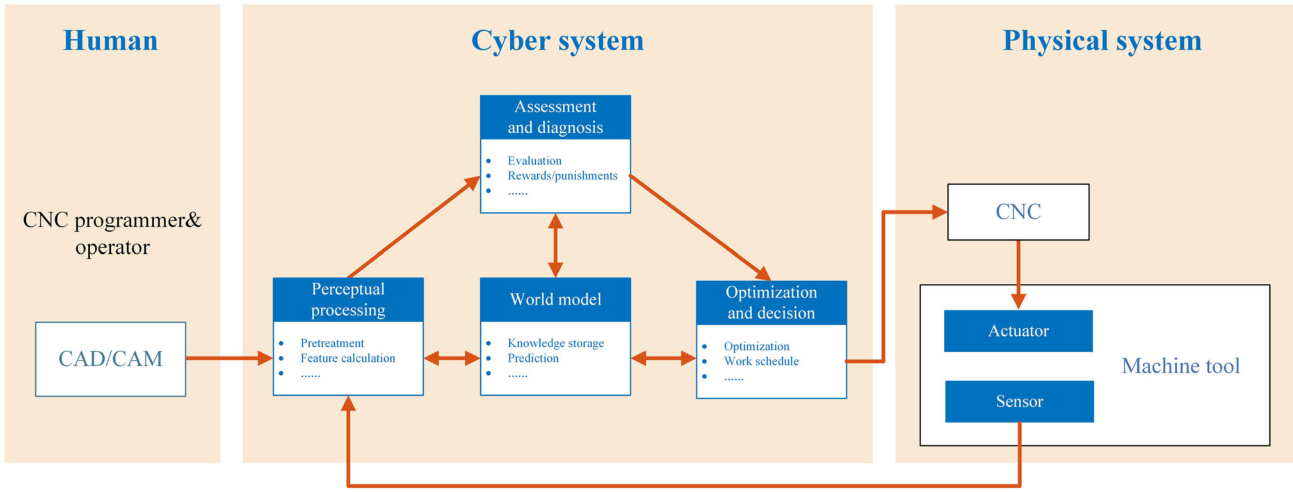



**Fig. 6** HCPS model and intelligent elements of intelligent CNC system

work plan and presents the final execution result in the environment.

- *Sensor* The sensor module mainly provides input for the cyber system, and the data comes from the sensors deployed by the machine tool and the internal control data of the CNC system.

The model shows that the input to the cyber system comes from two parts: human participation and sensors in the physical system collect the other. The idea of intellectualization and digitalization can reduce human participation in repetitive and dangerous work. However, it is not to separate people from production activities but to achieve a good integration of people and technology and to leverage their respective advantages. Intellectualization improves the efficiency and quality of production activities by utilizing advanced technologies and systems, reducing errors and resource waste. Digitization involves transforming data and information into digital form for better storage, processing, and analysis. These are consistent with the original intention of our research and will be introduced in detail later.

Combining the model constructed above in Fig. 6, a hierarchical intelligent CNC system architecture is proposed according to the CNC system tasks and application requirements. As shown in Fig. 7, this system is divided into three layers.

- *Equipment layer* This layer comprises field equipment in the CNC workshop, including execution equipment and sensing equipment, such as CNC controller, PLC, sensor, servo equipment, bearing, machine tool and cutting tools, which are the objects that generate input to the system.
- *Cyber layer* This layer processes data and provides services to the upper layer in the form of interfaces, including five sub-layers. Data acquisition mainly deals with structured data (such as the workshop ERP system or enterprise database data), semi-structured data (such as network file data), and unstructured data (such as sensor data and CNC machining process data). Data transmission include common industrial field data transmission protocols. The platform support mainly includes the current popular big data and AI algorithm support platform, providing support for upper data processing. CPS unit focuses on realizing the core intelligent elements in the cyber system mentioned above. Data services focus on the algorithm logic and adopted performance. For example, task data with high real-time performance can be processed via edge computing. Intelligent algorithms include the underlying provided algorithm support and enterprise algorithm library. The meaning of the arrow in the CPS unit indicates the interaction and sharing of data between modules of the same type in the CPS unit, providing support for the diversification of applications.
- *Application layer* This layer is close to the user and presents various work requirements and application scenarios of the CNC system, such as CNC program optimization, tool life prediction, tool trajectory planning, real-time machine monitoring, and fault diagnosis, providing users with intelligent management services.

## Modeling process and digital solution for path optimization

Previous CNC architectures tended to focus on the physical world (Liu et al., 2019a). The establishment of HCPS model in the CNC system is conducive to the interactive integration of physical entities and digital entities to better realize machining path optimization and real-time feedback during

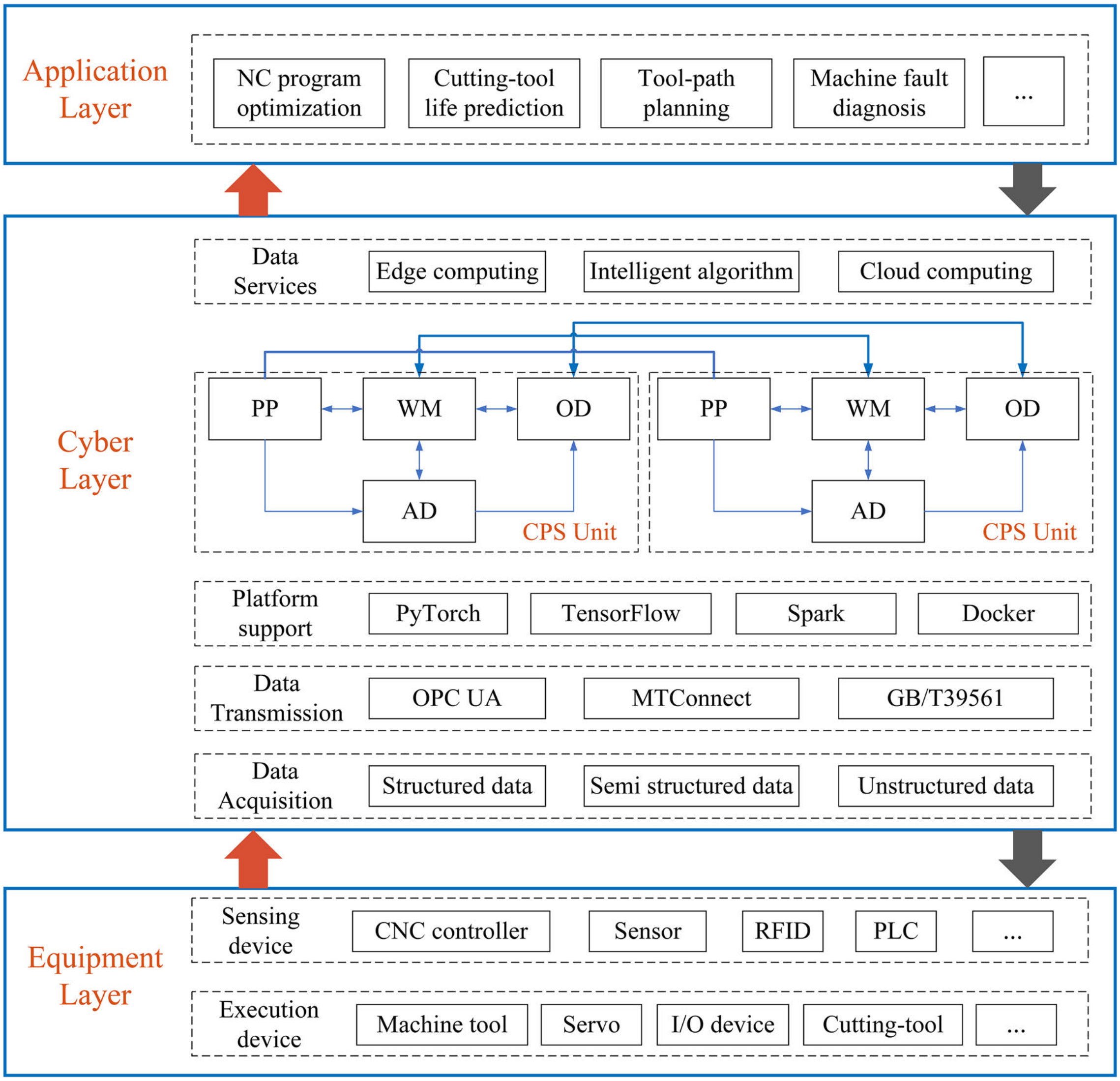



**Fig. 7** Three-layer architecture of intelligent CNC system

the machining process. The modeling process is shown in Fig. 8. Real-time information interaction between the CPS unit and the physical entity is realized through the digital thread. The intelligent elements in the CPS unit can be abstracted as intelligent agents. During the establishment of the HCPS model for the CNC system, the first step is to establish an information model for the input of the system, followed by constructing a mechanism algorithm for processing this information in order to inform the system of what information has been received and how to deal with it.

*Cyber agent* To better express the intellectual level of an intelligent CNC system, the cyber system is abstracted as an agent, and its intelligent degree is described from three factors (Meystel & Albus, 2000), and its model diagram is established as shown in Fig. 9.

- *Computing power* Describe the brain of the system, that is, the computer's computing power.
- *Algorithm* Describe the complexity and functionality of the system used to process input data, establish models, generate work plans, judge rewards and punishments, and real-time communication algorithms between modules.
- *Data* Describe the data stored by the system in its storage space and the value of the data.

The cyber agent has algorithms, computing power, data modules, and methods for interacting with other modules. This paper focuses on the application of algorithm module in tool path optimization. In the algorithm module, algorithms provided by shared machine learning (such as

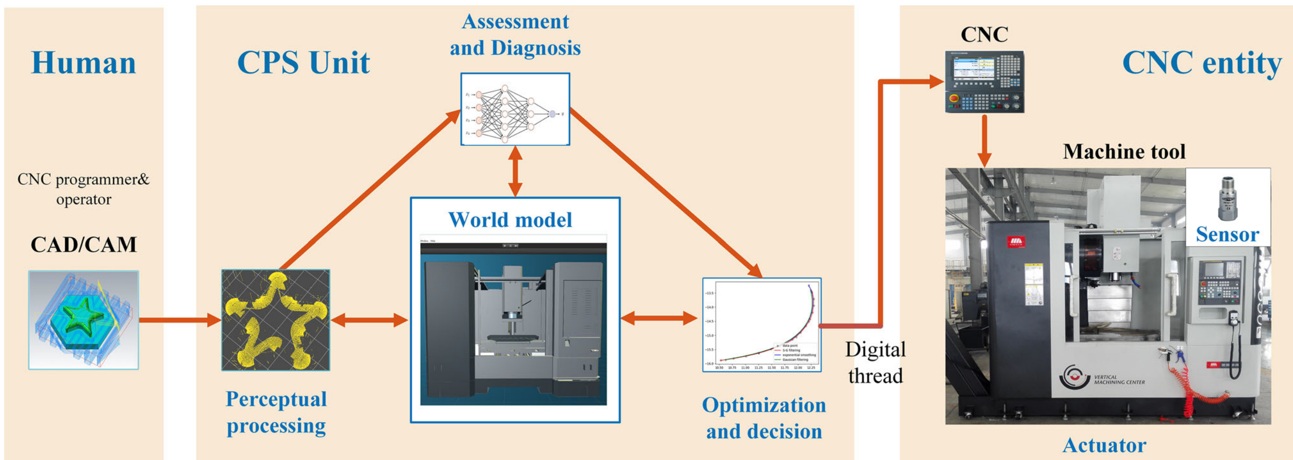



**Fig. 8** The HCPS modeling process of intelligent CNC system

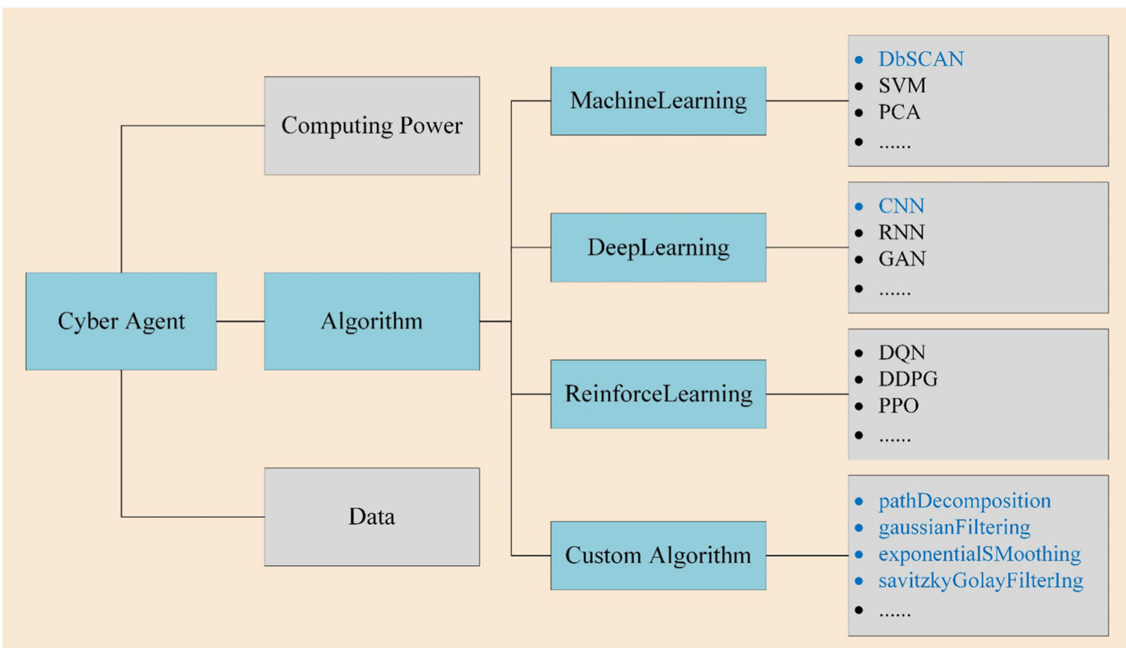



**Fig. 9** Structure of the Cyber agent model

sklearn), deep learning (such as PyTorch), and reinforcement learning frameworks or algorithms implemented by self-coding. Provide algorithm support for tool path optimization or other intelligent functions, thereby helping to construct a mechanical model for handling CNC system tasks. All algorithms in the cyber agent in this paper are implemented using Python 3.9, with PyTorch2.0.0 being used for deep learning, scipy1.7.1 being used for computing in machine learning and gym0.21.0 being used for reinforcement learning.

In current CNC information acquisition technology, because different CNC system manufacturers use their own standards, there is a lack of compatibility between communication ports and protocols, which makes the integration of CNC system equipment insufficient to improve the cost of data acquisition, thus hindering information sharing between field equipment. Emerging digital threading technology (such as OPC UA and MTConnect) has become increasingly favored due to its advantages of real-time operation, accuracy, standardization, etc. MTConnect can maintain consistency in the data interaction of CNC equipment through standardization. OPC UA uses an object-oriented address space to build an information model, supporting multiple data types such as attributes, features, methods, and events (Bruckner et al., 2019). Considering the compatibility of the middleware researched and developed in the previous stage and the information model established with OPC UA, as well

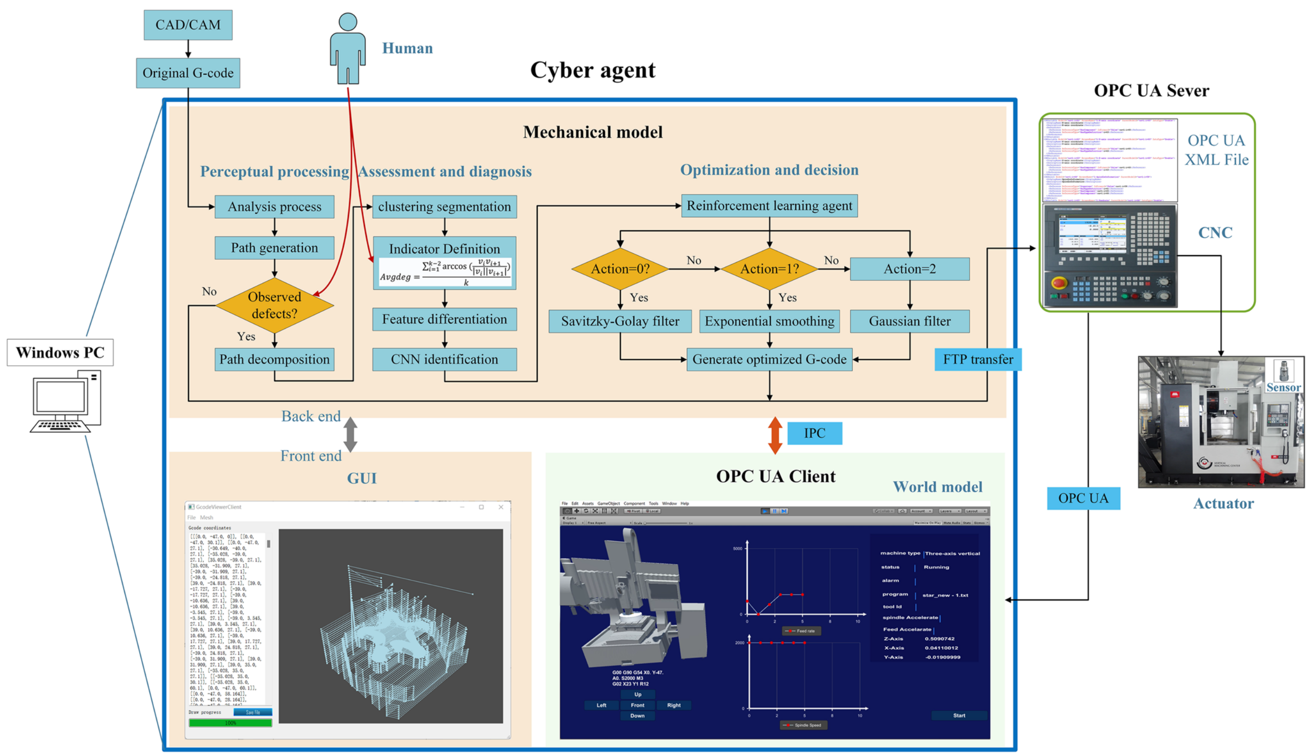



Fig. 10 Details of the technical solution for tool path optimization

as the convenience OPC UA provides for the user to customize the model in the industrial field, the research work in this paper uses OPC UA for information model establishment and data transmission. The specific technical solution is shown in Fig. 10.

The original G-code file is input into the cyber agent by the human operator, where it undergoes perception processing, evaluation and diagnosis, and optimization and decision to form an optimized G-code file. This optimized file is then transmitted to the CNC system via the file transfer protocol (FTP), which executes machining instructions based on it. Simultaneously, the feed axis position and spindle speed information are communicated in real-time to the cyber agent for digital twin machining simulation via OPC UA using a publish/subscribe model. Within the CNC system, a custom dynamic-link library data reading function binds with the OPC UA node data source, enabling real-time data collection and transmission. The current tool path information can be transmitted to the digital twin application for visual path generation and display at any stage of perception processing, evaluation and diagnosis, or optimization and decision. Inter-process communication (IPC) and event handling mechanisms are used here. When new data is read from the output stream of the monitored process, an event is triggered, and the event processing function is called. In this case, human operators serve a supervisory role in the perception processing module, using their expertise and experience to define criteria within the evaluation and diagnosis module. The information model of the tool path is described in "Information modeling of machining paths" section. The perception processing, evaluation and diagnosis, optimization and decision modules all contribute to creating a path optimization mechanism model. Firstly, the perception processing module completes the analysis of process documents, path generation, and decomposition process based on intuitive human judgment to decide the necessity of subsequent processes. Secondly, in the evaluation and diagnosis module, the process of clustering segmentation and feature recognition is completed, and the defect elements of the path are graded by defining the formula that describes the fluctuation characteristics of the path. Finally, adaptive path optimization based on reinforcement learning algorithms combined with feature level is performed in the optimization and decision module.

The development process of the machine tool digital twin model used in this paper was established using AutoCAD (2019) software based on the detailed dimensions of the machine tool drawings and rendered using Unity3D (2021) software. Object components belonging to different feed axes were bound in Unity3D, and motion C# scripts were added. At the same time, the OPC UA client was integrated with the C# script to read the CNC system tool path information in subscription mode, and implement monitoring and simulation of the machining process. To facilitate a more detailed

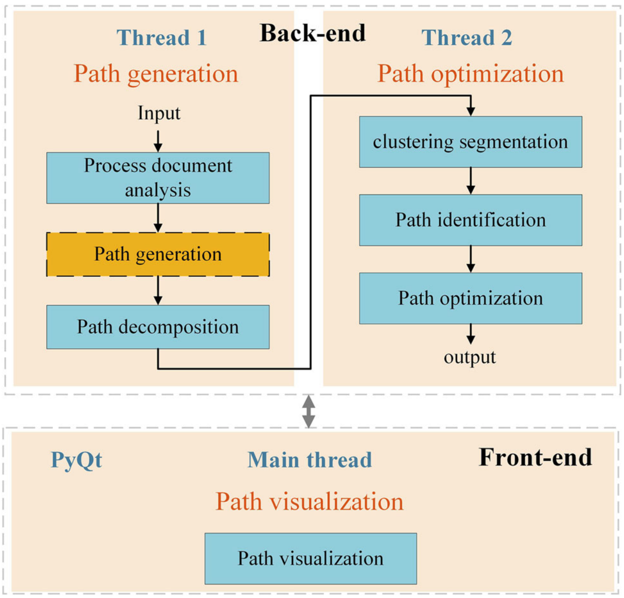



**Fig. 11** Multithreaded structure of path optimization application

observation of tool paths, a graphical user interface (GUI) for the cyber application in "Cyber application" section was developed. Once the digital twin model receives path information, the GUI client in the enabled state can synchronously read and display the tool path.

The advantages of implementing this HCPS framework for tool path optimization are as follows:

1. Early defect detection: After process analysis, the tool path information can be visualized for path generation and simulated processing based on digital twinning. Defects on the path can be detected in advance for CAD/CAM redesign or subsequent program optimization.
2. Optimization process automation: Realize G-code correction through program automation without requiring much manual labor from process personnel.
3. Multiple perspectives of tool path display: Visualization of point cloud images and trajectory images enables machining optimization during the production planning stage and displays process information such as spindle speed and feed rate during simulated machining, which helps in defect localization.

## Pipeline working mode of path optimization

As shown in Fig. 11, cyber applications are divided into front-end UI and back-end logic processing . The visualization client divides the path optimization process into path generation, path optimization, and path visualization threads. The visualization client acts as the main thread, providing a visual window for generating and rendering tool paths. The path generation process in Thread 1 includes process file analysis, path visualization generation, and path decomposition. After path decomposition, several path fragments are obtained. The path fragments that need to be optimized undergo clustering segmentation, path feature grading, and adaptive path optimization during the path optimization process of Thread 2. After completing human work in the preliminary experiment, the above process can form a fixed process and run in an assembly line manner.

In the process analysis stage, the back end of the system first analyzes G-code and auxiliary code (including M and T code) and other processing process files; analyzes the value information implied in the process files; and obtains the interpolation method, feed rate, spindle speed, and other related information. In this process, the position information of the feed axis under each path can be obtained, and the path and point cloud image can be generated using the position information, and the machining process can be decomposed into a series of machining paths. The position information of the feed axis is temporally related and is stored as position vectors in the form of sequential data in the application developed in this paper. In the path optimization process, the path optimization algorithm proposed later in this paper is used to optimize the timing sequence data of the feed axis position generated by the path, and the optimized position information is fed back to the process file for relevant optimization. The system's front end provides real-time feedback to the user on path generation and optimization through a visualization window. When machining complex structural parts, the machining process file will contain a large number of machining instructions, and the process analysis and path optimization process will produce a large delay, so it is difficult to realize real-time visualization of the machining path using the serial method.

As shown in Fig. 12, the improved front-end pipeline structure handles path segments in turn through three threads. When the pipeline of the optimization process is completed, the processes of path generation, path optimization, and real-time visualization are executed in parallel, and the corresponding processing cycle of each path fragment can output the visualization results for path optimization, which can greatly improve the real-time performance of the visualization process and facilitate operators to monitor and debug the optimization process in real time. Figure 13 shows an example of a multithreaded pipeline path optimization process.

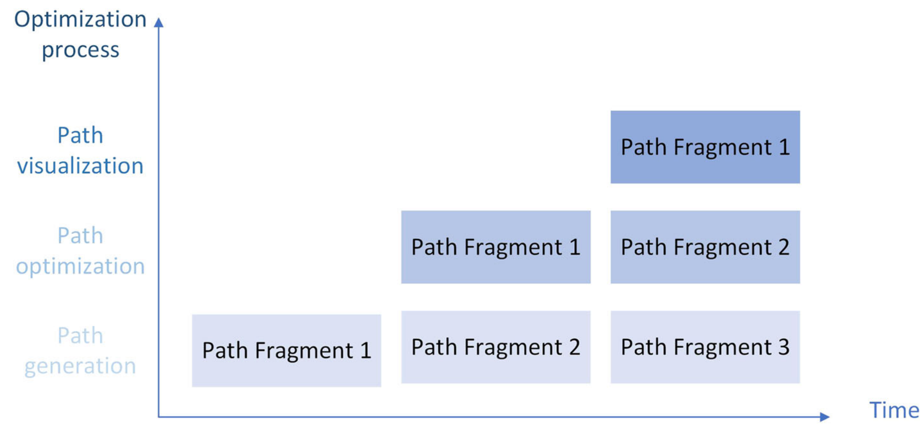



**Fig. 12** Schematic diagram of multithreaded pipeline processing path fragments

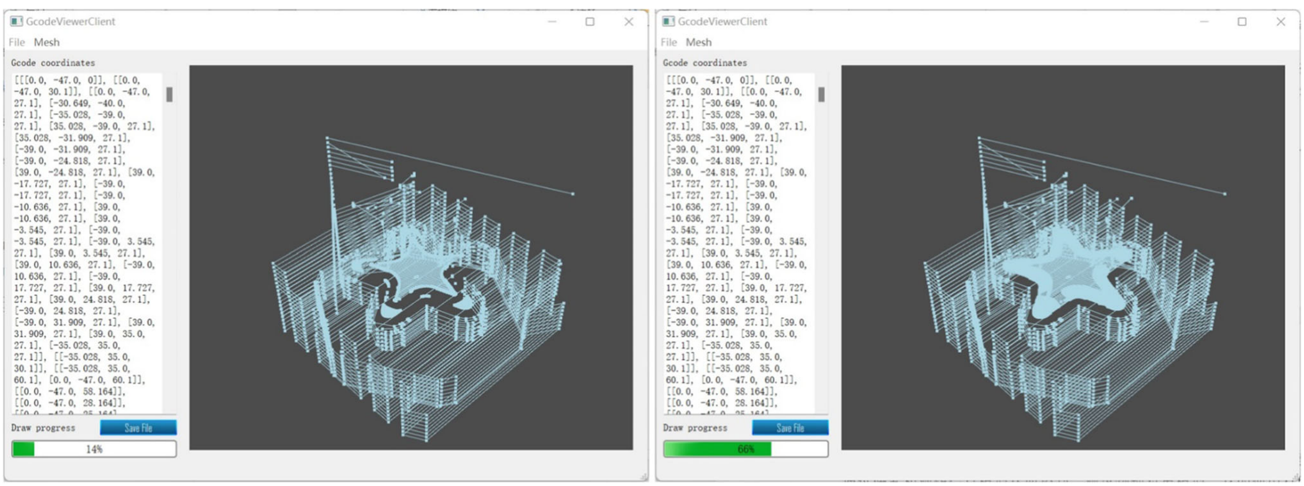



**Fig. 13** Visualization example during the path optimization process

## Experimental study

The machining of CNC systems is a very complex process. The tool path generated by a CAM system is usually a piecewise linear path with only one continuous position, which will lead to frequent deceleration of the CNC system and affect machining efficiency (Lyu et al., 2020). A rough tool path, uneven adjacent path, and slot points on the path will cause machining defects. Customizing and perfecting G-code on the machine must be practiced, and it is easier and more efficient to refine G-code directly than to edit the CAM tool path and then repost it. However, in today's business environment, it is not practical to train and maintain a manual programming operator with deep experience. Therefore, more advanced or convenient technical means are needed to identify, correct, and eliminate the negative path feature points generated in CNC program processing.

Feature point recognition is a classic pattern recognition problem, which can be generally divided into two types: the traditional manual method and deep learning-based method. For handmade methods, feature points can be identified by customizing rules describing intrinsic or extrinsic statistical properties intended to reflect the points. The current research based on deep learning methods includes methods using image vision (Hu et al., 2022) and methods based on point cloud features (He et al., 2023; Zhang et al., 2022). Image vision methods require a large amount of visual datasets collection and annotation work, and the demand for high computational efficiency also negatively impacts their dissemination in the market (Prunella et al., 2023). The method of using point cloud features also faces some challenges (Bello, et al., 2020).

Given the above problems, this section takes the numerical control system model based on CPS as an example, establishes the digital optimization process of the machining path, and optimizes the CNC program through the proposed trajectory decomposition and optimization algorithm.

### Establishment of experiment

The physical environment, digital twin model, and working process of the experiment are shown in Fig. 14. Here, the

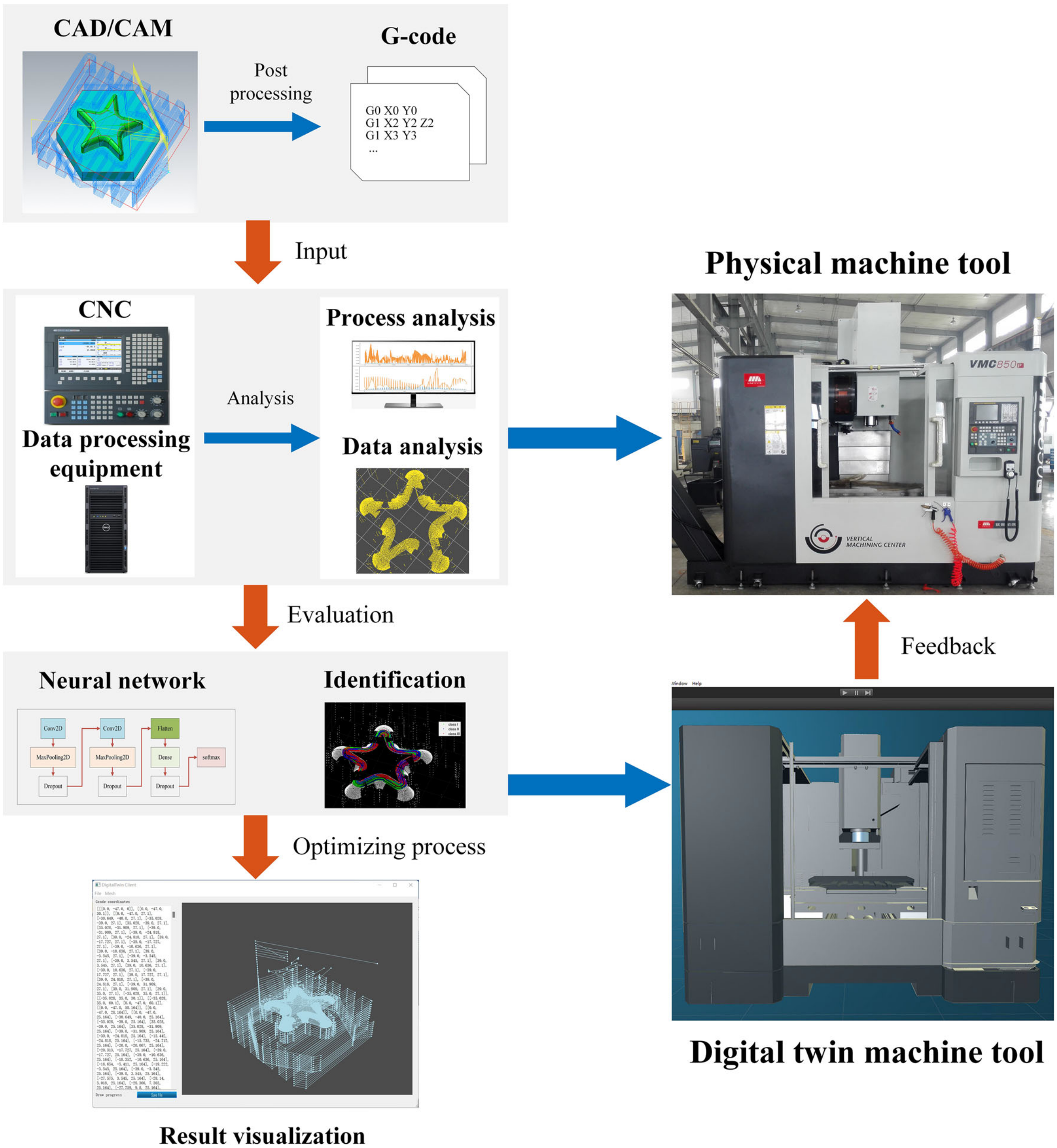



**Fig. 14** Experimental environment and associated workflow

CNC controller adopts a Golding GJ430 CNC system, and the data collector is a host with the OPC UA service, equipped with an Intel i9 11900 K CPU and an NVIDIA GeForce RTX 3090 GPU.

The basic flow of the work in Fig. 14 is as follows: Firstly, the CAD/CAM model is post-processed to generate the CNC process file, which are processed by the data collector program to parse the process files, complete the generation of tool paths and preliminary defect path analysis and perform hierarchical decomposition of the paths. Secondly, deep learning is used to identify defective paths. Finally, the paths are optimized using an adaptive path optimization algorithm based on reinforcement learning to generate visualization results on the client side and simulate machining in the digital twin model to form feedback to the CNC system.

## Information modeling of machining paths

The information model referred to in "Modeling process and digital solution for path optimization" section is built in this step, as shown in Fig. 15. The machining path information is an essential part of the information in the CNC. The machining path information includes the path number, spindle information, and feed axis information, whereas the

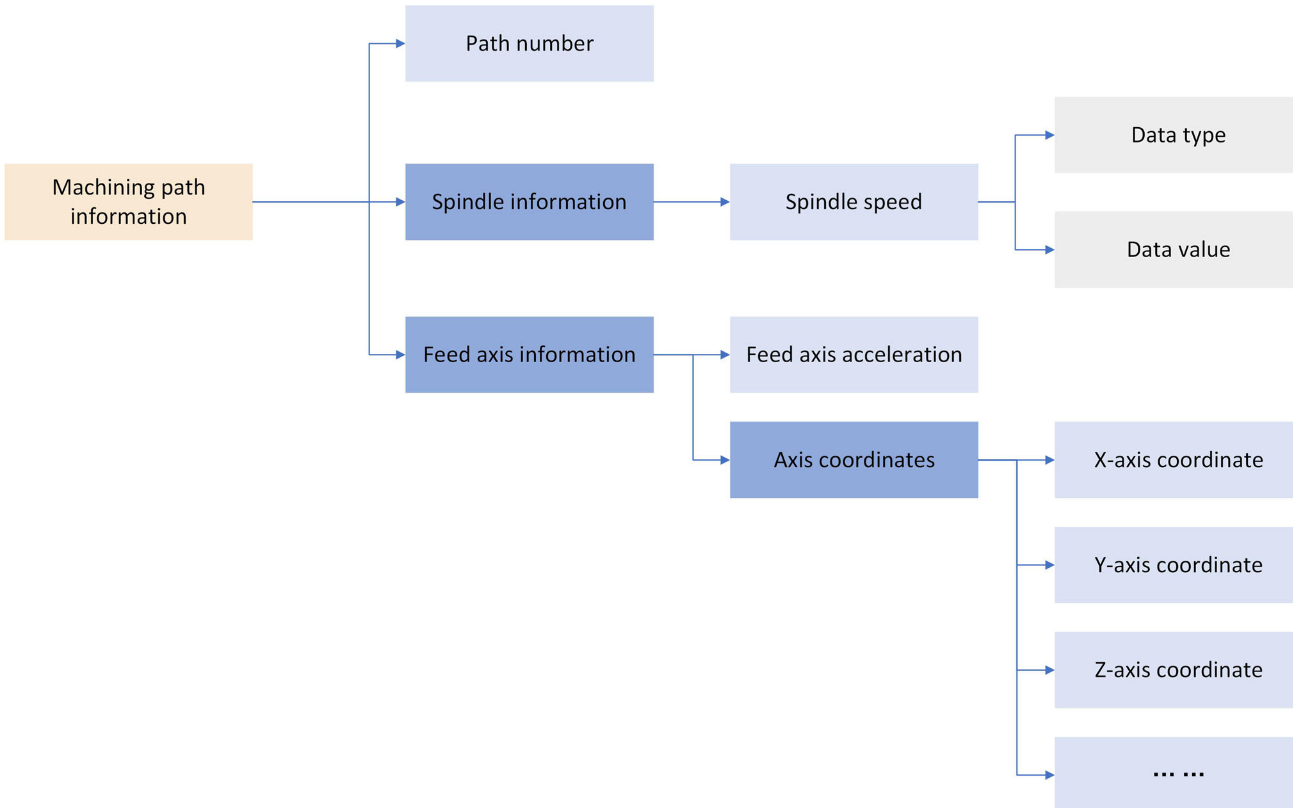



**Fig. 15** Structure of Machining path information model

feed axis information further contains the feed axis acceleration and coordinates. The above data contain attributes such as the index of the data item, readability, data type, and data value; some individual information also contains timestamp attributes. The path numbering data set in the diagram is weak sequential data, which supports the acceleration of the process flow.

OPC UA proposes a unified address space model to realize unified access and processing of information by providing a unified data interface. The server only needs to be called once to obtain a variety of service information, including current data collected by the OPC UA server, historical data, and subscribed data changes and events, which form the basis for the OPC UA server to realize its basic functions.

The creation and transmission of the information model is realized via OPC UA. The machining path information class is established and mapped to object nodes in the OPC UA address space, and each data item in the machining information class is mapped to variable nodes. Users can then access or modify the values of variable nodes in the address space through the OPC UA client.

## Mechanism of path optimization

### Analysis of process documents

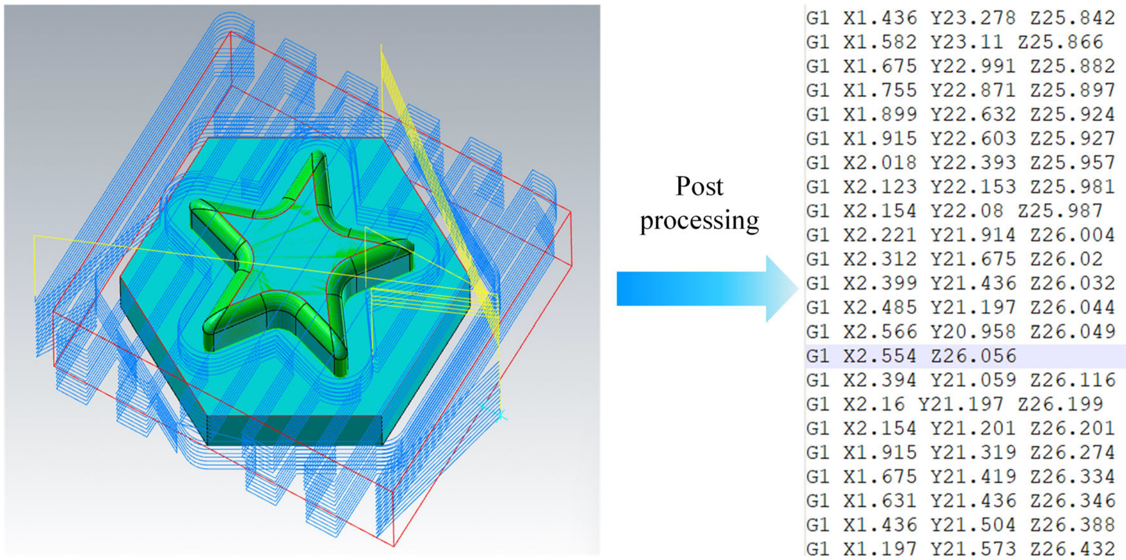
The high-speed and high-precision motion control of high-speed machining CNC machine tools is an important means to improve processing efficiency and quality. G-code is the most commonly used motion control language in CNC systems. The G-code used in this article complies with ISO 6983, one of the most commonly applied G-code standards.

Figure 16a presents a CAD model of a pentagram boss. A total of 16,225 lines of G-code are generated by this model, part of which is shown in Fig. 16b.

The analysis process for G-code mainly uses the following steps.

1. *Lexical analysis* This step mainly focuses on the lexical analysis of the following functions related to motor control:

(1) Function G-code: G00 (fast short-range movement), G01 (linear machining ), G02 (clockwise arc machin-




**(a)** Pentagram boss model **(b)** Partial G-code example

**Fig. 16** The pentagram boss CAM model (**a**) and the partial G-code example generated by it (**b**)

ing), G03 (counter-clockwise arc machining), G90 (absolute coordinates), G91 (relative coordinates), etc.

(2) Coordinate position: X, Y, and Z (three-dimensional space coordinates of the position points).
(3) Arc parameters: I, J, K (the relation between the position point in the arc motion and the center of the circle), and R (radius of the arc).
(4) Processing parameter: F (feed speed).
(5) By defining the keyword regular matching, the G-code, coordinate position, arc parameters, and other data are located and extracted.

2. *Data preservation* Save the parsed data for subsequent processing.

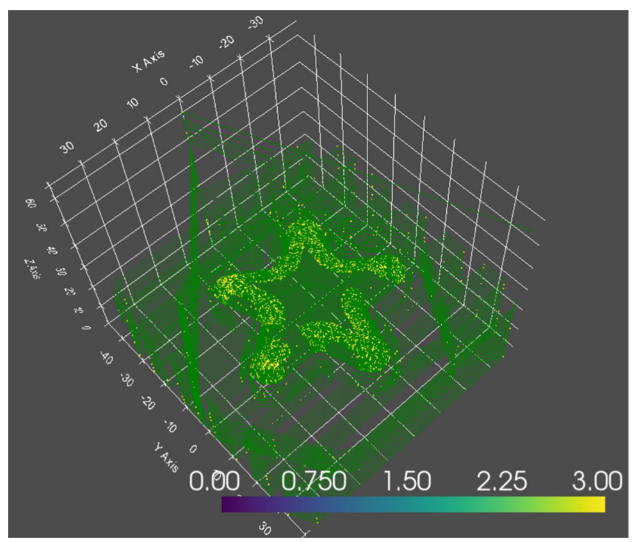



**Fig. 17** Path generation of the original G-code

### Path generation

1. Path generation

In the path generation stage, path generation and point cloud generation are carried out on the data files saved after process analysis. The point cloud has the characteristics of invariance in arrangement, translation, and rotation and can be locally amplified and expanded, which is convenient for process review and analysis. Through examination and analysis of the path and point cloud generation results, the path defects can be preliminarily determined. As Fig. 17 shows the path generation of the feed axis coordinate data, and Fig. 18 shows its corresponding point cloud generation and local amplification, it can be seen from Fig. 18 that there are more bad points and uneven adjacent paths in the original model of the pentagonal boss.

2. Path decomposition

To locate the defective path, it is necessary to analyze the global machining path to determine the area where the defect is concentrated. The principle of the path decomposition algorithm is as follows. Since the five-pointed star boss is an up-and-down structure, the overall trend of the tool path is a horizontal circular hierarchical path. Through a judgment plane perpendicular to the z-axis, and following the movement of the path point, the tool circular path is decomposed hierarchically, the transition point on the axis is found, and the entire complex path is decomposed into several relatively continuous parts. The exponential weighted average is then used to determine the decision plane to reduce the influence of outliers or jitter. The path hierarchical decomposition algorithm flow is shown in Algorithm 1. In the algorithm, $len()$ is a function to calculate the length of the data sequence, and $getPoint()$ is a function to add the data sequence to the set.

**Algorithm 1** Path decomposition algorithm

```
Input:
points: All the three-dimensional coordinates
axis: Divide the plane according to the direction of the coordinate axis
b: The distance threshold of the current point from the partition plane
min_n: The minimum number of point sets in a plane
alpha: The weight of the exponentially weighted average
Output:
blocks: List of path sets after partitioning
procedure
blocks ← [ ]
current_block ← [ ]
n ← 0
plane_position ← points[0][axis]
for each point in points do
  distance ← point[axis] - plane_position
  n ← n + 1
  if abs(distance) < b then
    current_block ← getPoint(current_block, point)
    n ← n + 1
    step_size ← alpha
    plane_position ← plane_position + step_size * distance
  else if len(current_block) >= min_n then
    blocks ← getPoint(blocks, current_block)
    current_block ← [point]
    plane_position ← point[axis]
    n ← 1
  end if
end for
if len(current_block) >= min_n then
  blocks ← getPoint(blocks, current_block)
end if
return blocks
```

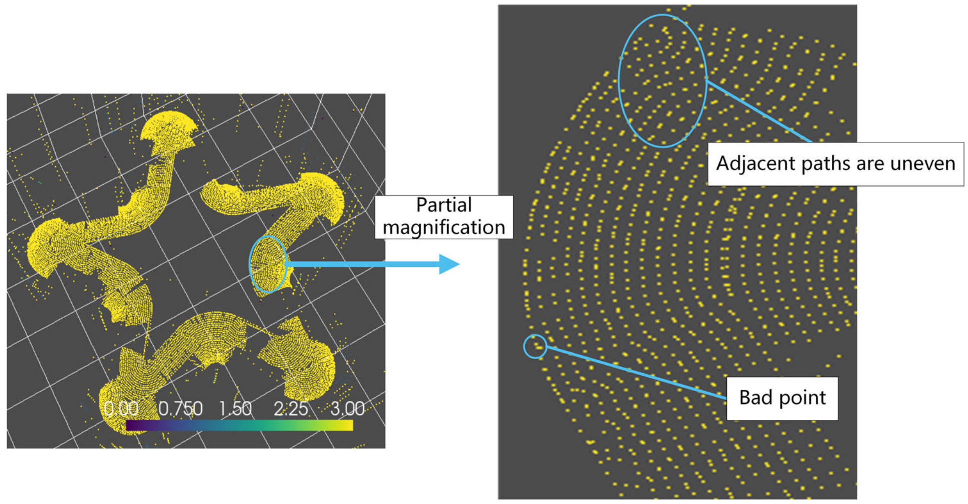
The parameter input values of the algorithm in the experiment in this paper are axis = 2, i.e., the hierarchical decomposition path of the z axis; b = 0.01; min_n = 10; and alpha = 0.1. The projection images of some paths on the z-axis are shown in Fig. 19. The defective locations based on the decomposed path segments are mainly concentrated in Fig. 19d, while the enlarged image is shown in Fig. 19e.

#### Path optimization

To improve the processing accuracy of the defect path, the defect path processed by the above steps (Fig. 19e) is segmented in a ring. The principle is to use an axis of the starting point as the reference line for segmentation, with the data sequence spanning two reference lines formatted as a ring.

The clustering algorithm based on Euclidean distance can automatically divide the points in the data set according to Euclidean distance to achieve segmentation. This division has a positive impact on the processing of spherical convex sets but is difficult to extend to arbitrary shapes. The density-based clustering method has strong self-adaptability and is good at finding clusters with irregular shapes. To further determine the local characteristics of the defect path and improve the adaptability of the optimization process, the density-based clustering method (DBSCAN) algorithm is used to perform clustering segmentation operations on

**Fig. 18** Point cloud and its partial magnification images generated from the original G-code



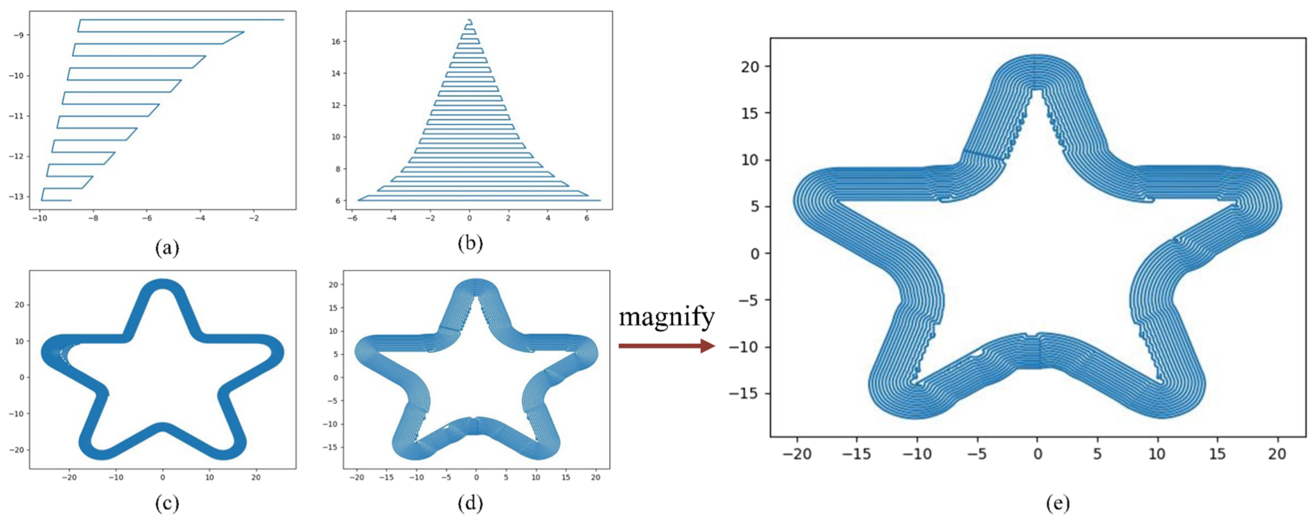




**Fig. 19** Example of a partial path fragment and enlargement of a defective path fragment, where **a** shows Fragment 24, **b** shows Fragment 27, **c** shows Fragment 36, **d** shows Fragment 35, and **e** is the enlargement of **d**

the annular star paths, setting the algorithm parameters scan radius equal to 2.5 and the minimum number of inclusion points equal to 2. Examples of some star shaped annular paths are shown in Fig. 20.

To distinguish the local characteristics of the intercluster pathpoint sets in the above clustering results, we use normal vector to describe and reflect the local variation characteristics of the point sets in the cluster. The normal vector is an important attribute of the scattered point cloud model, which can be used to describe the model feature information. If the path points in a cluster are relatively flat, the direction change of the normal vector of each point in the neighborhood is relatively flat. Conversely, when the fluctuation of path points within a cluster changes greatly, the direction of normal vectors of adjacent points also changes relatively greatly. Therefore, the corresponding path optimization method can be determined by estimating the direction change of normal vectors of the cluster.

To quantify this index and highlight the local characteristics of the point sequence, we calculate the normal vectors between adjacent points in each cluster in a neighborhood of size k and use the angle between adjacent normal vectors as the radian across adjacent points to reflect the flatness of the point sequence. The average cumulative value of the radian in the k neighborhood reflects the overall fluctuation of the data sequence in the k neighborhood.

Suppose that $p_1, p_2, \ldots, p_k (k > 2)$ is a sequence of data points and $v_1, v_2, \ldots, v_k$ is the normal vector of $\overrightarrow{p_1 p_2}$, $\overrightarrow{p_2 p_3}$, $\ldots$, $\overrightarrow{p_{k-1} p_k}$, then the index of the fluctuation degree of the data sequence in the k-neighborhood is

$$Avgdeg = \frac{\sum_{i=1}^{k-2} \arccos\left(\frac{v_i v_{i+1}}{|v_i||v_{i+1}|}\right)}{k} \quad (1)$$

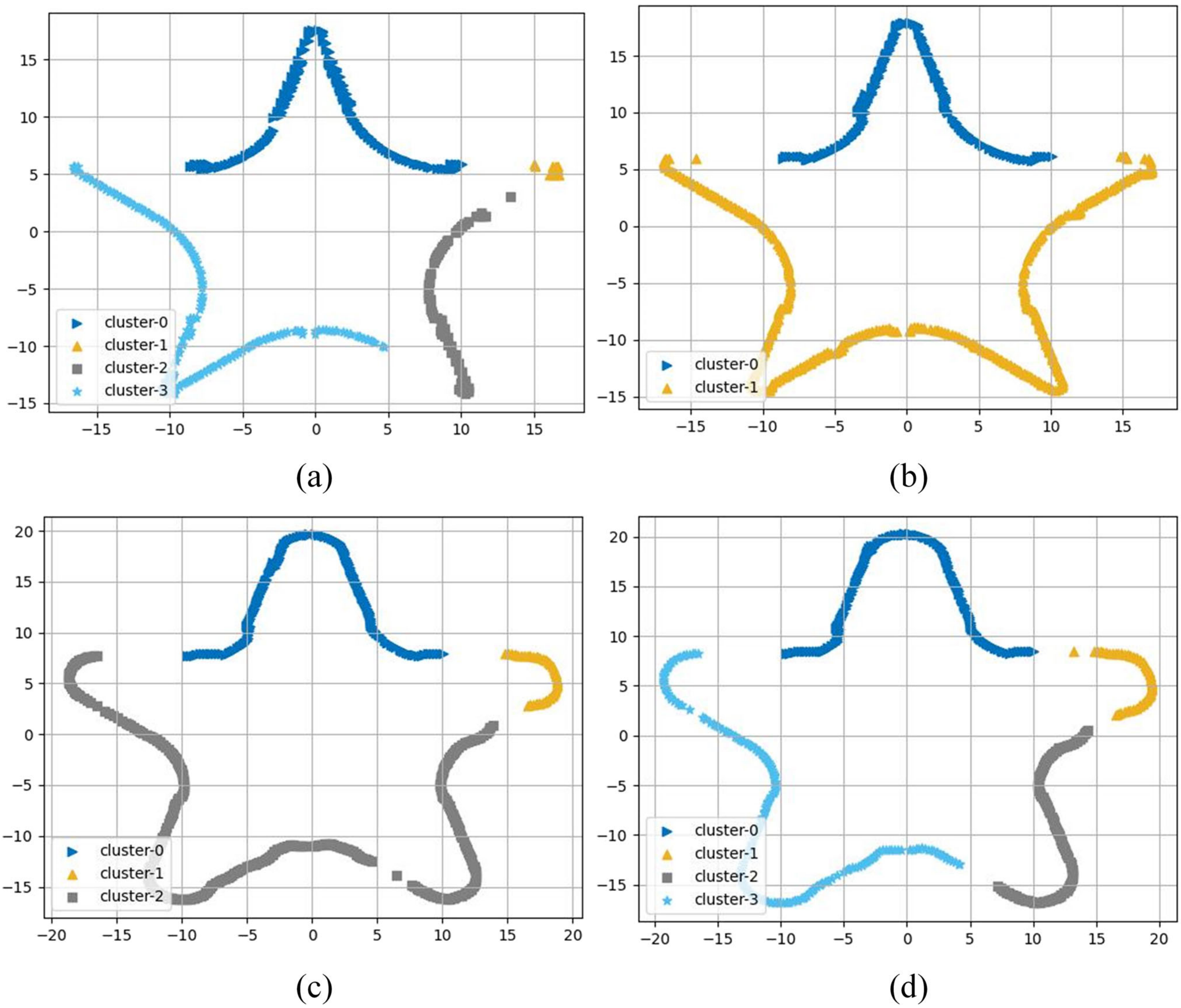



**Fig. 20** Partial examples of clustering results, where **a** shows the cluster results for Ring 1, **b** for Ring 2, **c** for Ring 8, and **d** for Ring 10

**Table 1** Statistical table of feature classification

| $Avg$ deg | Amount | Proportion (%) | Description | Grade of defect |
|---|---|---|---|---|
| Avgdeg $\geq$ 10.0 | 136 | 37.06 | The path is uneven and has many offset points, which has a great influence on the machining | III |
| Avgdeg $\in$ (3, 10) | 146 | 39.78 | The path defects are relatively serious, and some paths are not flat, which has an impact on machining | II |
| Avgdeg $\leq$ 3 | 85 | 23.16 | The path defects are light and have little impact on the machining | I |

The larger this value is, the more obvious the unevenness of the pathpoint set becomes. Based on the actual observations of the experiment, when k = 20, the distinction of features is obvious. Table 1 shows further statistical results, presenting the size and proportion of $Avgdeg$ in 367 fragments with length k, and describes the data characteristics. In addition, the defect level is established in the table for the fluctuation degree of the data sequence.

Based on the feature engineering described above, the features of the paths were identified and classified using deep learning methods to facilitate operator review of the process. The structure of the established deep learning network is shown in Fig. 21, and the input data are the first two columns of the above 367 three-dimensional data of length 20. The reason for omitting the third dimension is to consider that the data in the third column has slight variations, which do not contribute significantly to the features. Before the data is fed into the neural network, it is reshaped into 367 $\times$ 1 $\times$ 20 $\times$ 2, where 1 is the number of channels, after which the data is fed into the convolutional neural network for training in batches. The neural network has two convolutional layers using 32 and 64 2 $\times$ 2 convolutional kernels, respectively, and both of them use a sliding window with a step size of 1 and a 1 $\times$ 1 padding. The data sizes of the inputs and outputs of each layer are labeled in Fig. 21. The model uses an Adam optimizer with a learning rate of 0.001, a cross-entropy loss function, a training period of 1000, and a batch size of 10.

The training and testing accuracy of this classification algorithm and the receiver operating characteristic (ROC) curve are used as evaluation metrics to show the model's effectiveness. As shown in Fig. 22, where Fig. 22a illustrates the training and testing accuracy of this classification algorithm, and Fig. 22b illustrates the ROC curves of this three-classification model, the area under the ROC curve (AUC) values of the ROC curves of the model on the three categories are 0.90 for class 1, 0.97 for class 2, and 0.93 for class 3, which all show good differentiation ability. Figure 23 demonstrates the results of the algorithm's classification.

Based on the analysis and statistical results of the above feature sequences, it can be seen that it remains difficult for a single optimization scheme to have a good effect on all data sequences due to the differences in the characteristics of the data sequences in each k-neighborhood. Therefore, we adopted the corresponding filtering and smoothing algorithms for different defect features. The filtering and smoothing algorithms adopted are introduced in the following section.

- *Gaussian filter* Gaussian filter is a linear low-pass filter, which can effectively suppress noise and smooth data. Its essence is the weighted mean filter with weights, and the weight is related to the distance between the elements in the filter and the filter center. The weight value is the largest in the center of the convolution kernel and decreases around. Gaussian filters are very effective for suppressing noise that follows a normal distribution.
- *Second-order exponential smoothing* Exponential smoothing is a modification of the moving average algorithm that can be used to smooth data with short-term trends. The first-order exponential smoothing can be used when the data series has no noticeable trend change, and the second-order exponential smoothing is based on the first-order exponential smoothing. It is suitable for the data series with a certain linear trend.
- *Savitzky-Golay filter* The S-G filter is widely used in the smoothing and denoising processing of data sequences. The filtering method fits 2 m + 1 discrete data in a window by the least square method. The polynomial fitting of the local interval is realized by convolution so as to remove the noise signal in the data, highlight the characteristic information in the data, and reduce the influence of unnecessary information on the data. The most important characteristic of the S-G filter is to remove the signal noise while keeping the shape and width of the signal unchanged. The effect of the S-G smoothing filter is closely related to the width of the selected window, which can meet the needs of various occasions.

The various filter optimization parameter configurations in Fig. 24 are presented in Table 2.

Figure 24 shows the filtering optimization results of the partial k-neighborhood sequence data with different features. According to the results shown in Fig. 24a, the data sequence is uneven and relatively scattered. Additionally, the linear relationship or trend is not obvious, and the Gaussian filtering algorithm has a better effect. In Fig. 24b, the data sequence has a linear relationship or trend. Additionally, the convex—concave characteristics are not obvious, and the second-order exponential smoothing method has a better effect. In Fig. 24c, the data sequence is relatively flat with fewer fluctuations, and the S-G filter algorithm has better optimization and fitting effects on this kind of data. Accordingly, an adaptive filtering algorithm is proposed that utilizes the Double Deep Q-Networks (DDQN) algorithm (Van Hasselt et al., 2016) to select the filter automatically. Specifically, we would like to retain our findings (with a probability of 0.7) and give the reinforcement learning agent some autonomous decision-making power to assist us in achieving better-filtering results. In the optimization process, the decision-making of the reinforcement learning agent is achieved by setting a random number $sampled$ within [0,1) and a probability constant $prop$ with a value of 0.7, combined with the ε-greedy strategy, specifically, when $sampled$ is greater than $prop$, the filtering is performed according to the defect grade classified by $Avgdeg$ in formula (1); otherwise, the filtering is

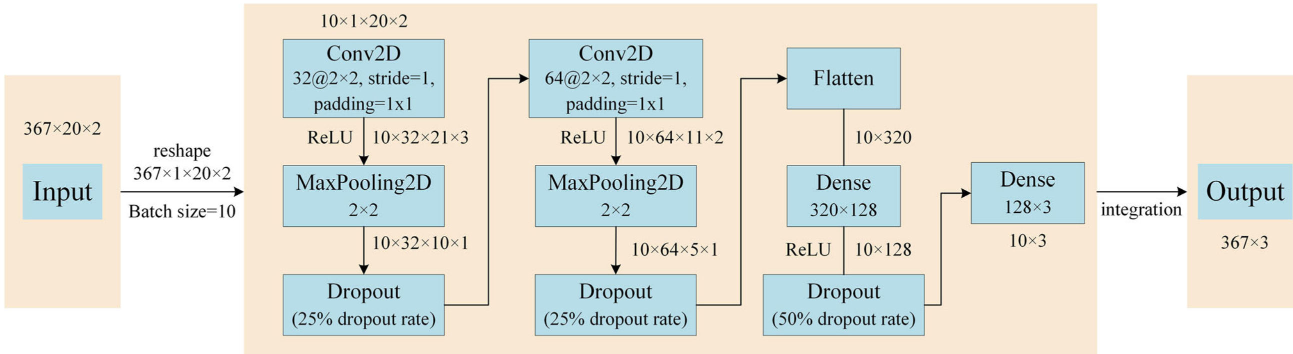
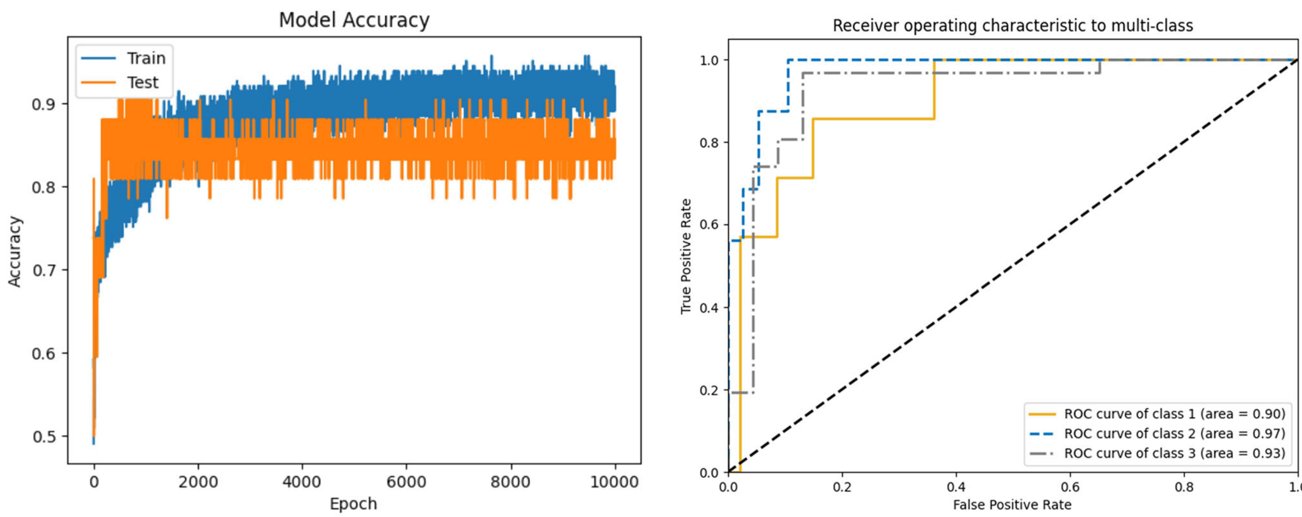



**Fig. 21** Network structure and processing flow of deep learning model




(a) The classification accuracy of the model

(b) ROC Curves for Multi-class Classification

**Fig. 22** Accuracy and ROC curves for the classification algorithm

performed based on the learning of the reinforcement learning agent and the ε -greedy strategy to choose the filtering method to use. Based on this idea, the reinforcement learning environment is defined as follows:

- *State(s)* A continuous block of 3D data of size k = 20.
- *Action(a)* There are three actions {0,1,2}, action 0 corresponds to S-G filtering, action 1 corresponds to exponential smoothing and action 2 corresponds to Gaussian filtering.
- *Reward(r)* Two factors are considered in the reward: the difference between the original and smoothed data and the consistency between successive smoothed blocks. Dynamic Time Warping (DTW) (Senin, 2008) is used to measure the above differences, assuming that $A = [a_1, a_2, \ldots, a_n]$ and $B = [b_1, b_2, \ldots, b_m]$ are two data sequences, where $n$ and $m$ are the lengths of A and B, respectively. DTW aims to find a path to minimize the two sequences' distance metric between them. Use the following dynamic

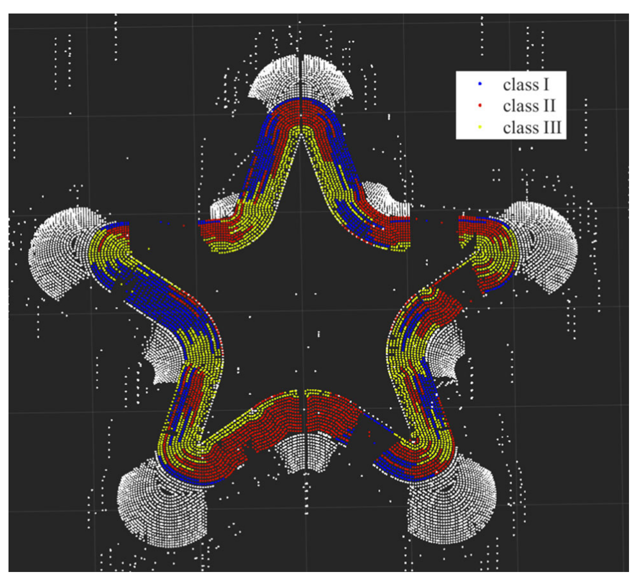
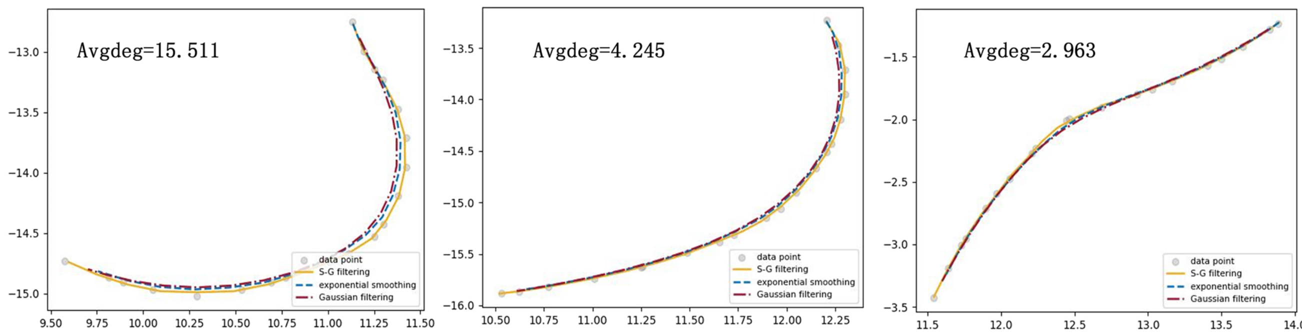



**Fig. 23** Visualization of point clouds for algorithmic classification results




(a) (b) (c)

**Fig. 24** Comparison of different filtering effects for three defect levels, where **a** shows the filtering effects for defect level III, **b** shows the filtering effects for defect level II, and **c** shows the filtering effects for defect level I

**Table 2** Parameter configuration in the experiment

| Filtering or smoothing method | Parameter configuration |
|---|---|
| Gaussian filter | $wsize = 9, \sigma = 5$ |
| Second-order exponential smoothing | $a = 0.3$ |
| Savitzky-Golay filter | $sgwsize = 7, \beta = 3$ |

programming Eq. (2) to find this path.

$$D(i, j) = d(a_i, b_j) + min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\} \tag{2}$$

where $d(a_i, b_j)$ is the Euclidean distance between $a_i$ and $b_j$ and $D(i, j)$ is the minimum distance from $A[1 : i]$ to $B[1 : j]$. The initial conditions are $D(0, j) = \infty$, $D(i, 0) = \infty$, $D(0, 0) = 0$. The final value of the DTW(A, B) distance is $D(n, m)$. Define the reward function as Eq. (3).

$$r(s, a) = -(DTW(s, S) + DTW(S_{prev}, S_{first})) \tag{3}$$

where $s$ is the current state and $S$ is the current data block after smoothing by applying action $a$, where $S_{prev}$ is the last data point of the previous smoothed block, and $S_{first}$ is the first data point of the current smoothed block. The former in the reward function represents the difference between the original and smoothed data based on the original data, and the latter indicates the consistency between successive smoothed blocks. Considering the difference in data magnitude between the two parts of the reward function (3), the maximum and minimum values of each part are maintained separately in the algorithm, and the Min–Max Normalization is performed when calculating the reward, and the normalized reward function is Eq. (4).

$$r(s, a) = -(DTW(s, S)_{norm} + DTW(S_{prev}, S_{first}))_{norm} \tag{4}$$

In reinforcement learning, the $Q$ function is used to represent the expected future reward that can be obtained by taking an action $a$ in a given state $s$, denoted as $Q(s, a)$. Its recursive definition is given in Eq. (5).

$$Q(s, a) = r + \gamma max_{a'} Q(s', a') \tag{5}$$

where $r$ is the immediate reward obtained by taking action $a$ in state $s$, $\gamma$ is a discount factor that determines the importance of future reward, $s'$ is the next state after taking action $a$, and $max_{a'} Q(s', a')$ is the maximum $Q$ value of all possible actions in state $s'$. In practice, the $Q$ function is usually approximated by neural networks. In the DDQN algorithm used in this paper, two identical neural networks are used, divided into a policy network and a target network, where the policy network is used to select actions, and the target network is used to evaluate the value of these actions. This separation helps to minimize the overestimation of $Q$ values. In training, the policy network with parameter $\theta_t$ is first used to select an action a′ that maximizes the $Q$ value using Eq. (6).

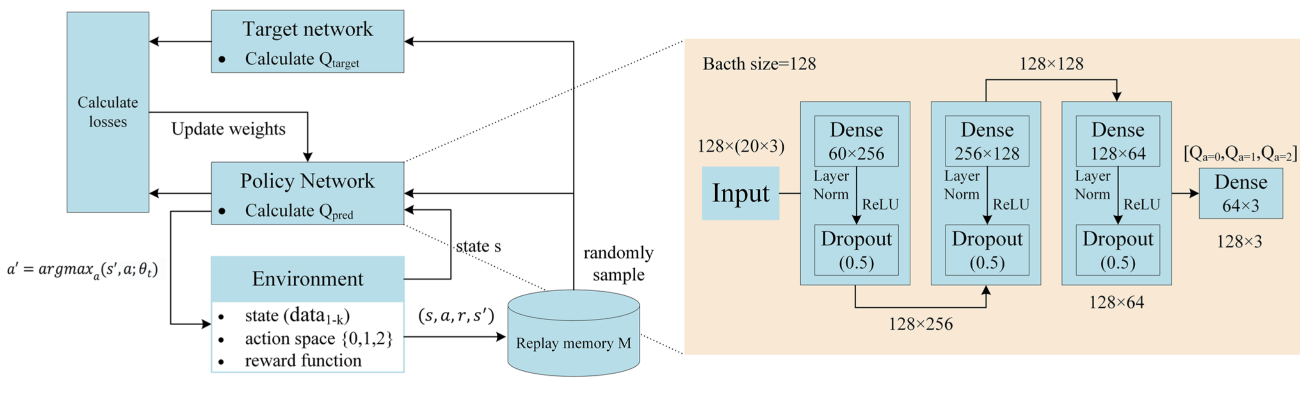
$$a' = argmax_a(s', a; \theta_t) \tag{6}$$

Next, the target network with parameter $\theta^-$ is used to evaluate the value of the action $a'$ chosen by the strategy network using Eq. (7):

$$Q_{max} = Q(s', a'; \theta^-) \tag{7}$$

The $Q_{max}$ is the maximum $Q$ value in state $s'$ to evaluate the current ε-greedy strategy. The target $Q$ value is then calculated by combining the immediate reward $r$ and the future $Q$ value using Eq. (8).

$$Q_{target} = r + \gamma Q_{max} = r + \gamma Q(s', argmax_a Q(s, a; \theta_t); \theta^-) \tag{8}$$

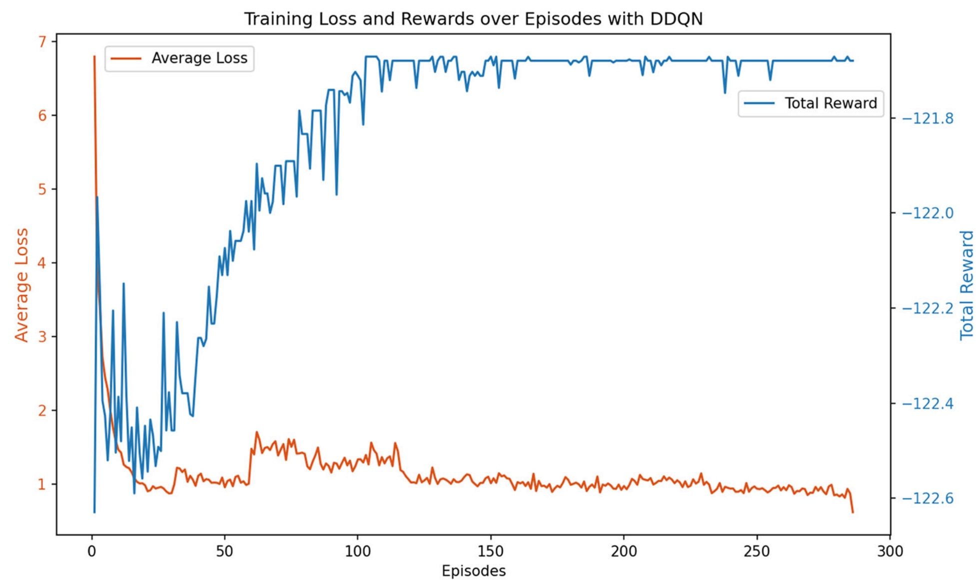



(a) DDQN structure and execution process

(b) DDQN network structure

**Fig. 25** DDQN execution process (**a**) and detailed network structure (**b**)

**Fig. 26** Total reward change per step in DDQN training and average loss change per batch in optimization




Finally, the policy network calculates the predicted $Q$ value for a given state and action using Eq. (9):

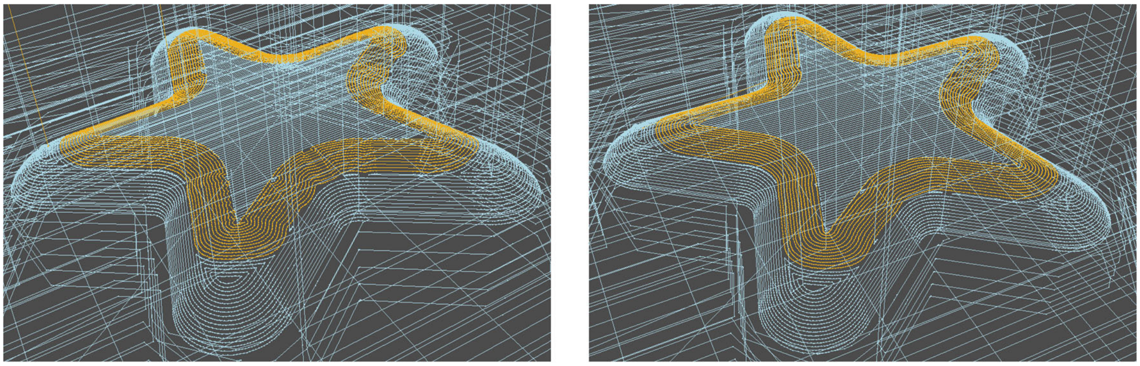
$$Q_{pred} = Q(s, a; \theta_t) \tag{9}$$

In this way, the difference between $Q_{target}$ and $Q_{pred}$ can be used to update the parameters of the policy network, thus gradually approximating the actual Q function. This process is schematically shown in Fig. 25, where Fig. 25a demonstrates the general operation of DDQN learning, and Fig. 25b gives details of the policy network structure, which is structurally identical to the target network, with the dimensional sizes of the inputs and outputs of each layer labeled in Fig. 25b.

The overall algorithmic flow and hyperparameter settings are shown in Algorithm 2, and the parameter configurations of the filters in Algorithm 2 are the same as in Table 2:

**Algorithm 2** Adaptive Filtering Algorithm Based on DDQN

**Input:**
$Data \leftarrow \{Data_1, Data_2, \ldots, Data_n\}$: Imported 3D raw data set
$k \leftarrow 20$: Size of neighborhood
$lr \leftarrow 0.001$: Learning rate
C ← 10000: Replay memory capacity
$\gamma \leftarrow 0.99$: Discount factor
EPS_START ← 0.9, EPS_END ← 0.05, EPS_DECAY ← 200: $\varepsilon$-greedy strategy parameters
BATCH_SIZE ← 128: Number of experience samples used in each optimization step
TARGET_UPDATE ← 10: Frequency of target network update
num_episodes ← 300: Total number of cycles of training
prop ← 0.7: Probability of filtering based on $Avgdeg$
**Output:**
$Data'$: The set of optimized data sequences
**Procedure**
Initialize replay memory M to capacity C
Compute $avgdeg$ for each $k$ neighborhood according to formula (1), save in list $Avgdeg$
Initialize environment $env$, policy network $policy\_net$, target network $target\_net$
$steps\_done \leftarrow 0$, $Data' \leftarrow []$
**for** $episode \leftarrow 1$ to $num\_episodes$ **do**
  Initialize state $s \leftarrow \{Data_1, Data_2, \ldots, Data_k\}$, $done \leftarrow$ False, $chunk\_idx \leftarrow 0$
  **while** not $done$ **do**
    $avgdeg \leftarrow Avgdeg[chunk\_idx]$
    $sample \leftarrow$ random number from [0, 1)
    **if** $sample <$ prop **then**
      **if** $avgdeg > 10$ **then**
        action $a \leftarrow 0$
      **else if** $3 \le avgdeg \le 10$ **then**
        $a \leftarrow 1$
      **else**
        $a \leftarrow 2$
      **end if**
    **else**
      $\varepsilon \leftarrow$ (EPS_START - EPS_END) * $exp$(-steps_done / EPS_DECAY) + EPS_END
      **if** $sample > (\varepsilon +$ prop$)$ **then**
        $a \leftarrow$ Action selected based on formula (6)
      **else**
        $a \leftarrow$ random choice from {0, 1, 2}
      **end if**
    **end if**
    Execute action $a$, calculate reward $r$ according to equation (4), get next state $s'$, done
    Get $smoothed\_data$ based on current state $s$ and selected action $a$
    Update replay memory with the tuple $(s, a, r, s')$
    $s \leftarrow s'$, $steps\_done \leftarrow steps\_done + 1$, $chunk\_idx \leftarrow chunk\_idx + 1$
  **end while**
  Randomly extract experience of size BATCH_SIZE from replay memory M as B
  **for each** $(s, a, r, s')$ in B **do**
    Use formula (6) with $policy\_net$ to compute the action $a'$ that maximizes the Q value
    Use the $target\_net$ to compute the $Q_{max}$ value for next_state using equation (7)
    Set the target Q value $Q_{target}$ using formula (8)
    Compute the predicted Q value of $policy\_net$ $Q_{pred}$ using formula (9)
    Compute loss: $L \leftarrow \sum(Q_{pred} - Q_{target})^2$
    Optimize $policy\_net$ using $RMS\_prop$ optimizer to reduce loss $L$
    Update $policy\_net$ weights: $\theta_t \leftarrow \theta_t - lr * \nabla L$, $\nabla$ is gradient of $L$ with respect to $\theta_t$
  **end for**
  Update $target\_net$ every TARGET_UPDATE time steps
  Add $smoothed\_data$ to the end of $Data'$
**end for**
**return** $Data'$



(a) The model before optimization (b) The model after optimization

**Fig. 27** Visual comparison of the overall model before and after optimization

Figure 26 shows the variation of the average loss per BATCH_SIZE of optimization steps and the total reward per step throughout DDQN training. As can be seen from the figure, the average loss per BATCH_SIZE of optimization steps gradually decreases as the number of training steps increases. In contrast, the total reward per step gradually increases. This indicates that the model is learning to smooth the data better and that the chosen smoothing strategy is becoming more accurate.

The model before and after applying Algorithm 2 optimization is shown in Fig. 27, and the local details and coordinate values before and after optimization are shown in Fig. 28. As shown in Fig. 27, the optimized paths' smoothness and the adjacent paths' consistency are significantly improved compared to before optimization, which is more clearly illustrated by the local details in Fig. 28.

To evaluate the efficiency of the multithreaded pipelined work mode described in "Pipeline working mode of path optimization" section, we measured the execution time in both serial and pipelined manners 100 times, as illustrated in Fig. 29. The efficiency improvement was calculated by taking the difference between the mean execution time in serial mode and that in pipelined mode, and then dividing this difference by the mean execution time of the serial mode. Our results indicate that the pipelined mode improved the execution efficiency by 18.602% compared to the serial mode.

To measure the difference in toolpaths before and after optimization, define $AvgDiff$ in Eq. (10) as the average smoothness difference between each class of data before and after optimization, counting the data of each class before optimization (BO) and after optimization (AO), respectively, where n is the total number of each class, k is the size of the neighborhood. $Avgdeg$ is the smoothing degree defined by Eq. (1).

$$AvgDiff = \frac{1}{n}\sum_{i=1}^{n}\frac{BO[i]_{Avgdeg} - AO[i]_{Avgdeg}}{k} \tag{10}$$

The statistical results show that the $AvgDiff$ for categories I, II, and III are 4.006%, 13.191%, and 39.490%, respectively. It indicates that the average smoothness of each of the three classes has been improved to different degrees. The improvement is smaller for class I, moderate for class II, and more significant for class III.

## Cyber application

In this paper, the GUI of the Cyber application client was developed through the PyQt and PyVista library. PyVista is a helper module for the Visualization Toolkit (VTK), a Python toolkit that inherits the VTK data format and is compatible with VTK data objects. This package provides a complete set of interfaces and documentation that facilitates rapid prototyping, analysis, and visual integration of spatial reference datasets.

The visualization client interacts with the OPC UA server, and the OPC UA server is integrated in the GJ430 CNC system to realize the real-time transmission and visualization of machining data. The server completes the initial configuration by setting the port number and security certificate. The analyzed and optimized path information is mapped based on the path information model of the CNC system and transmitted to the OPC UA server to generate process files, which are used by the CNC system for processing.

The cyber client of this paper is shown in Fig. 30 and includes two main areas. The process file information and specific content can be viewed in area 1, while area 2 shows the visual point cloud image of the axis coordinates in the process file and the path image. The client back-end program analyzes and optimizes the machining path information immediately and feeds back the improved G-code to region 1. The operator can then read the historical process file stored locally through the operation bar above area 1 to load the content and obtain the display.

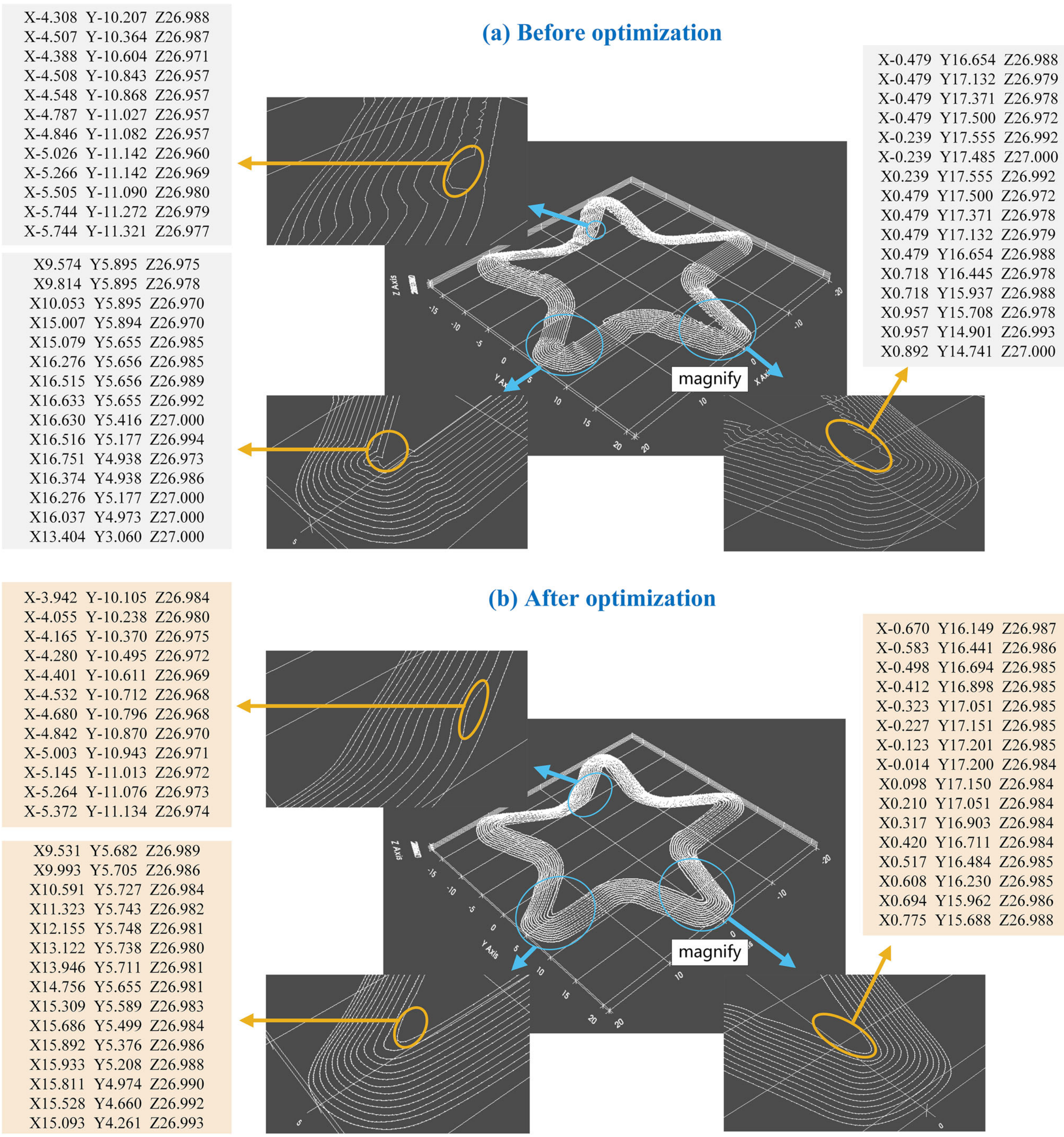



**Fig. 28** Comparison of local details of tool paths and G-code coordinates before and after optimization

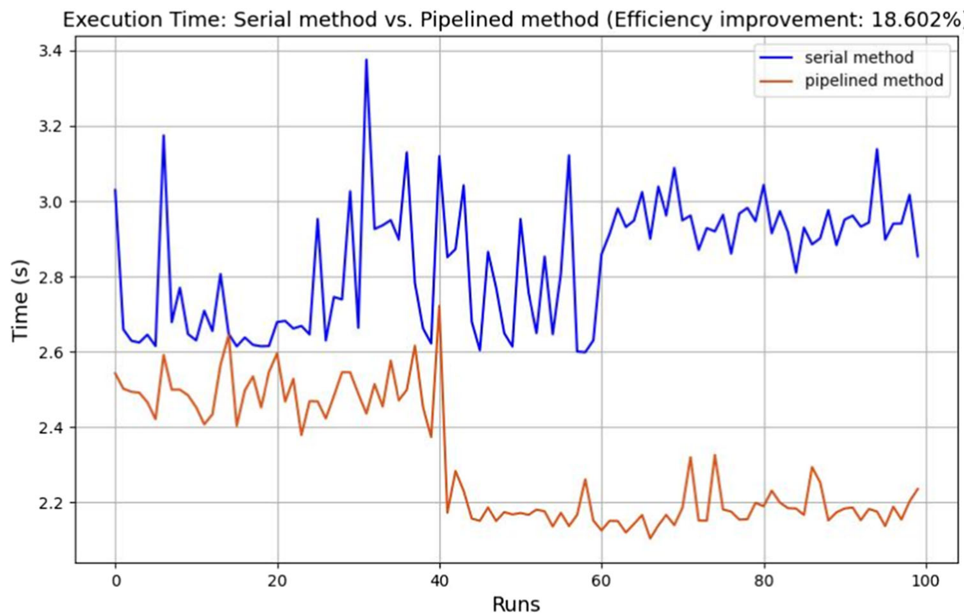



**Fig. 29** Comparison of 100 execution times between serial and pipelined methods

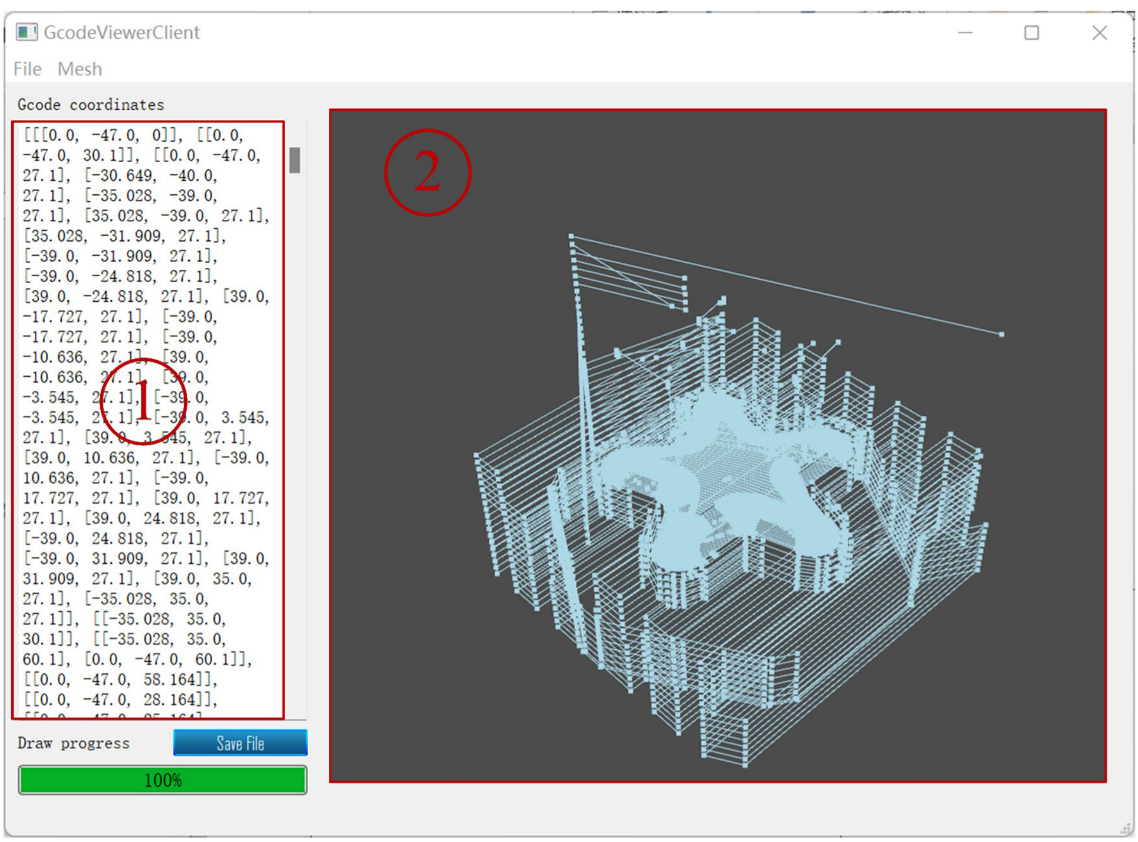



**Fig. 30** Cyber application GUI and functional areas

Through this client, the processing data between the CNC system entity and the cyber application can be exchanged in real time, which improves the quality and reliability of CNC system processing. At the same time, the technical threshold for the manual programming operator is reduced, and the operator does not need to modify the G-code manually to improve the machining effect, which greatly reduces the production costs.

## Conclusion

The path optimization of CNC system machining is one of the most effective ways to improve the machining quality and reliability of machine tools. This paper researches HCPS-based CNC system machining path optimization methods and put forward corresponding solutions for the problems of fragmentation of the design and manufacturing process, imperfect unified integration model, machining defects in CNC programs generated by CAM systems, and difficulties in the real-time analysis of the machining process. The main contributions of this research are as follows.

First, we analyzed the shortcomings of the previous data transmission mode between the CAM and CNC systems and established a closed loop of data transmission between the CAM system and machine tool controller using a new working mode. Secondly, we analyzed the motivation of CNC system intellectualization, proposed a scheme for the intelligence of the CNC system and the corresponding abstract model, and established the architecture of the HCPS-based CNC system. On this basis, the modeling process of the HCPS-based intelligent CNC system is given, the structure model and cyber agent model are established, and a digital solution for machining path optimization is proposed. Then, a set of pipelined path optimization processes and corresponding cyber applications were developed to improve the efficiency of path optimization and the real-time performance of the process analysis. Finally, the principle of the process analysis and the proposed path hierarchical decomposition algorithm, path generation process, and path optimization process were introduced through an application example. In the process of path optimization, the clustering algorithm of path segmentation, the deep learning classification algorithm for path defect features, and the adaptive filtering algorithm of path optimization were given, and the established Cyber application is shown at the end of the paper.

In this paper, based on the concept of HCPS, a digital solution for CNC machining path optimization was proposed, and the corresponding cyber application was established. However, the optimization results in this paper still have room for optimization when interpolation is carried out in the CNC. In the future, the research content of this paper will be expanded to a deeper level, and the trajectory planning method and interpolation optimization method for the pathpoint sequence will be investigated and integrated into the cyber application outlined in the present study.

**Author contributions** LZ: Conceptualization, Methodology, Data analysis and Writing. HY: Methodology, Software and Formal analysis. CW: Resources and Data curation. YH: Validation. WH: Software. DY: Supervision, Project administration.

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**Data availability** The data underlying this paper will be shared on reasonable request to the corresponding author.

## Declarations

**Competing interests** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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